



Diffusion Model: Application, Method, and Implementation

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Outline

- **Implementation**
 - **Image synthesis**
 - **ADM (classifier guidance, model ablation)**
 - LDM (latent diffusion model)
 - DiT (pure transformer model)
 - Classifier-free guidance
 - Image-to-image translation
 - Task-specific image restoration and enhancement
 - Zero-shot image restoration
 - Size-agnostic image generation
 - Video generation
 - Diffusion acceleration
 - Appendices for equations



DMs beat GAN on image synthesis [NIPS'21]

- Find better U-Net architecture through a series of ablations
 - Focus on 25-250 generation steps (25 w/ DDIM)
 - Scale up to 512x512 images
- Achieve better quality than GAN for
 - Unconditional image synthesis
 - Conditional synthesis with classifier guidance

$$\hat{\epsilon}(x_t) := \underbrace{\epsilon_{\theta}(x_t)}_{\text{Original noise estimator}} - \underbrace{\sqrt{1 - \bar{\alpha}_t}}_{\sigma_t} \underbrace{\nabla_{x_t} \log p_{\phi}(y|x_t)}_{\substack{\text{Gradient from} \\ \text{trained classifier} \\ \text{on noisy } x_t}} \times \underbrace{s}_{\text{Scaling factor}}$$

Samples with guidance

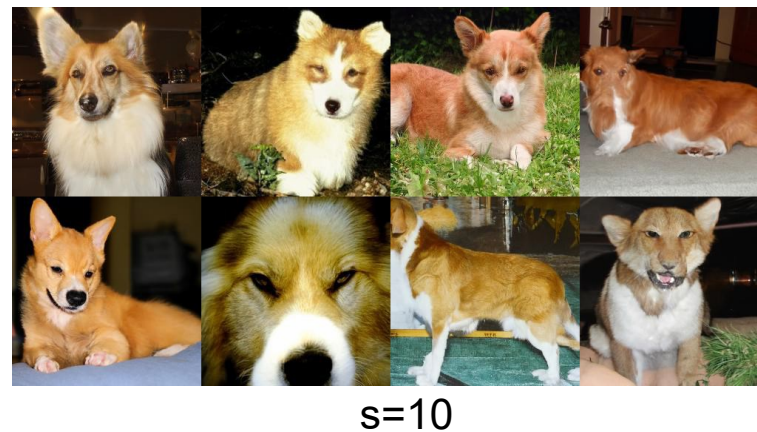
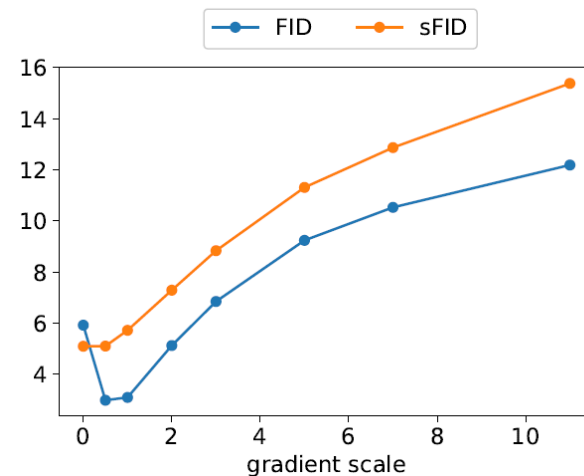
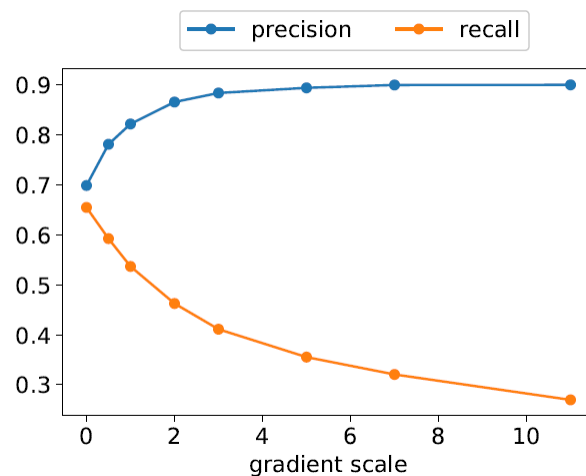
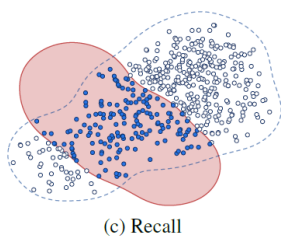
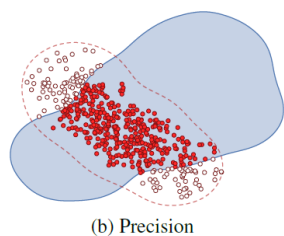


BigGAN-deep

Ablated Diffusion Model (ADM)

Trade-off with scaling factor s

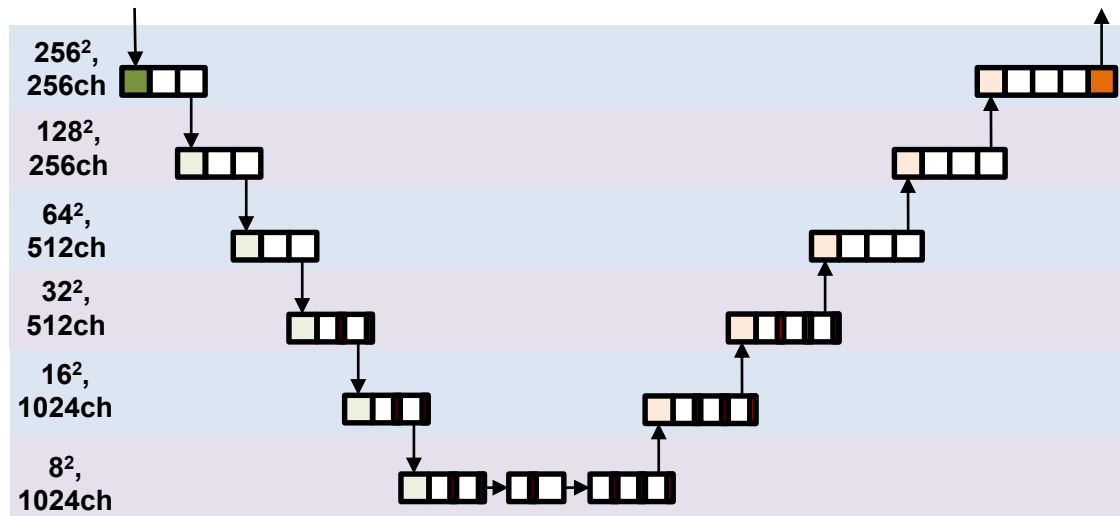
- $s \uparrow \Rightarrow$ fidelity (precision) $\uparrow$, diversity (recall) $\downarrow$





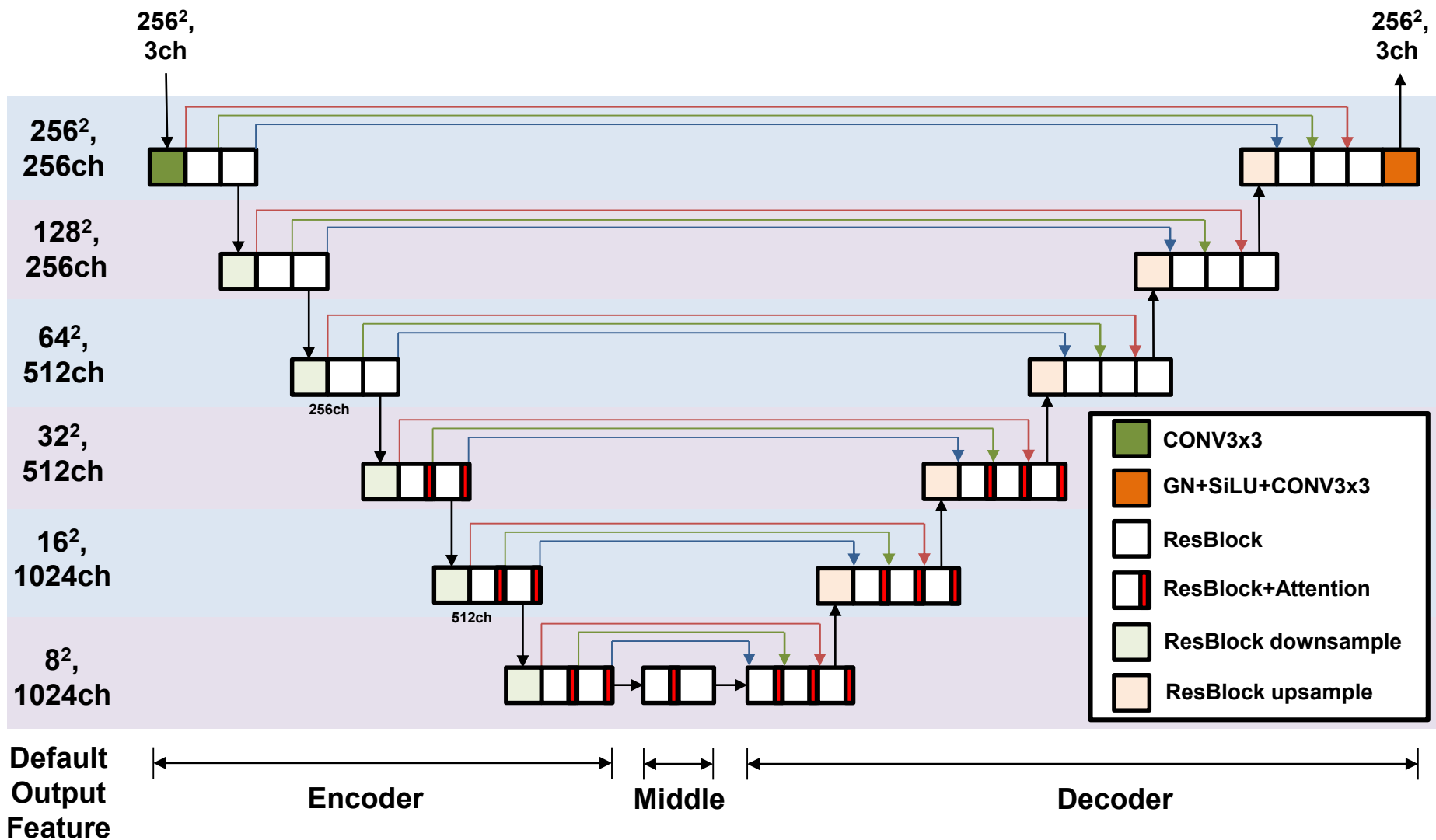
U-Net example: ADM for ImageNet 256

- Configuration
 - Base channel: 256
 - Level: 6; channel multiple: 1, 1, 2, 2, 4, 4
 - Number of ResBlock per level: 2 (informative)
 - Attention resolution: 32, 16, 8; Head channel #: 64
 - Model size: 554M; Complexity: 2.2TFLOPs per step



U-Net

(ADM for ImageNet 256)

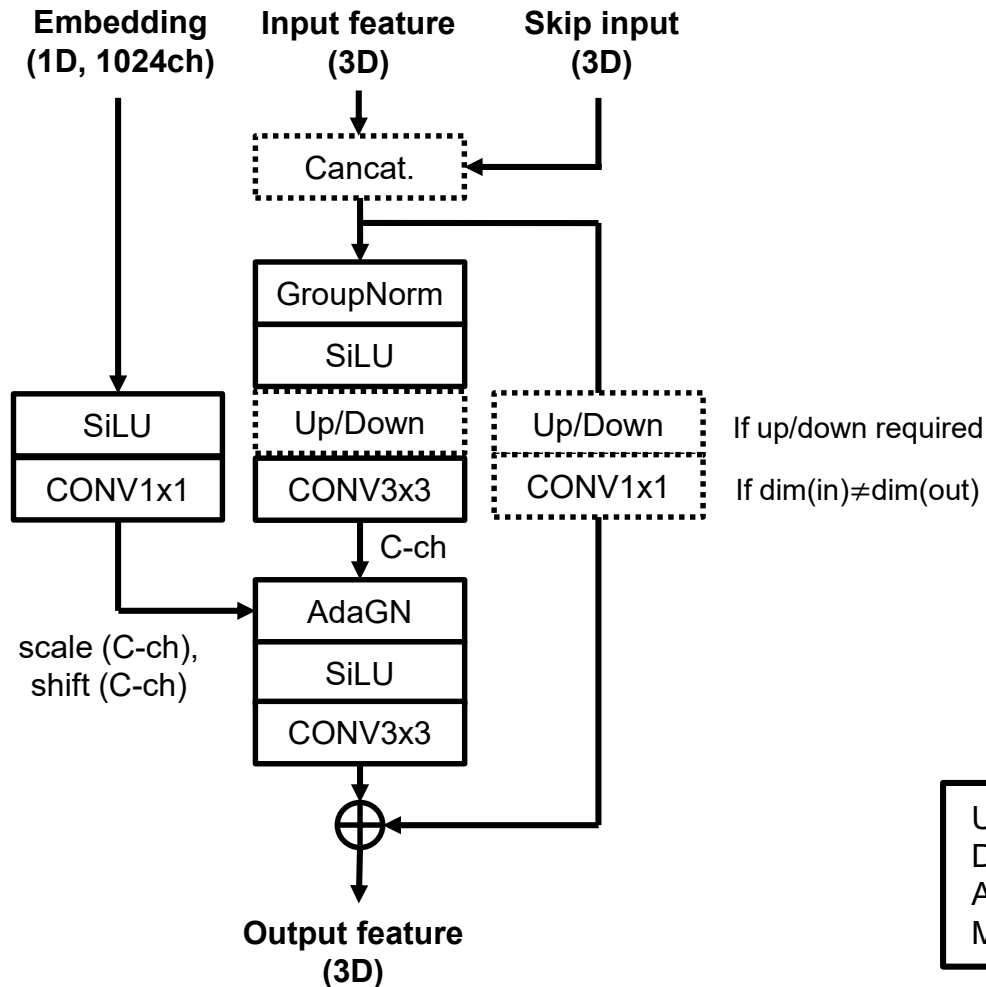




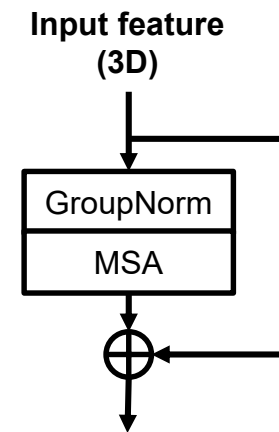
Building blocks

(ADM for ImageNet 256)

ResBlock:



Attention:

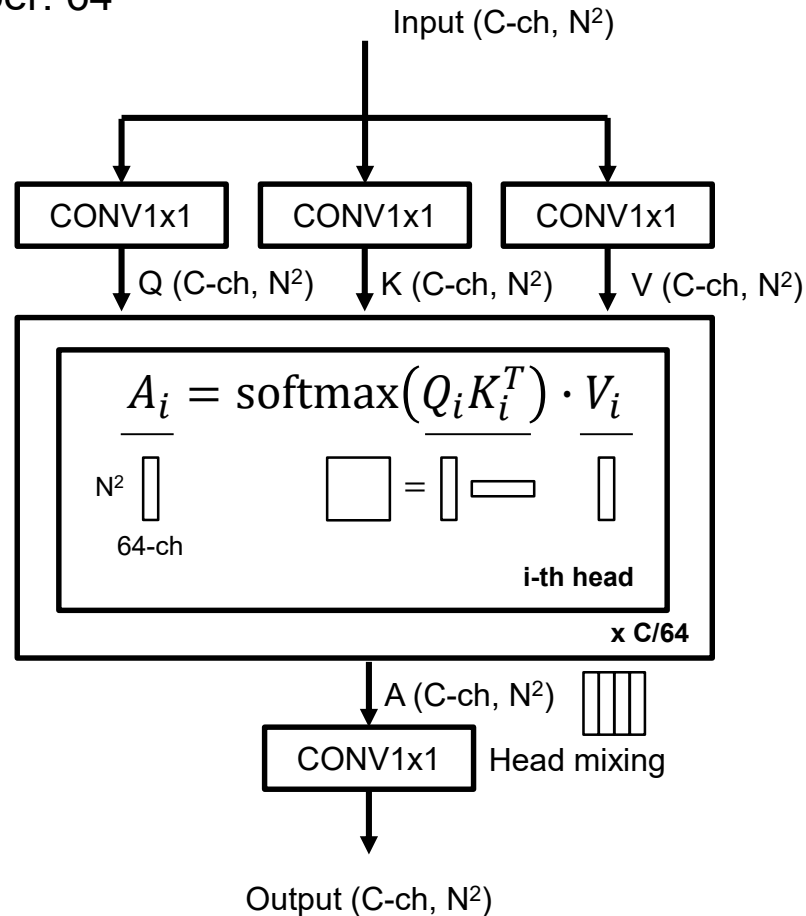


Up(x): Nearest upsample by 2
Down(x): Average pooling (stride 2)
AdaGN(x): $(1 + \text{scale}) \cdot \text{GN}(x) + \text{shift}$
MSA(x): Multi-head self-attention



Multi-head self-attention (MSA)

- Head channel number: 64





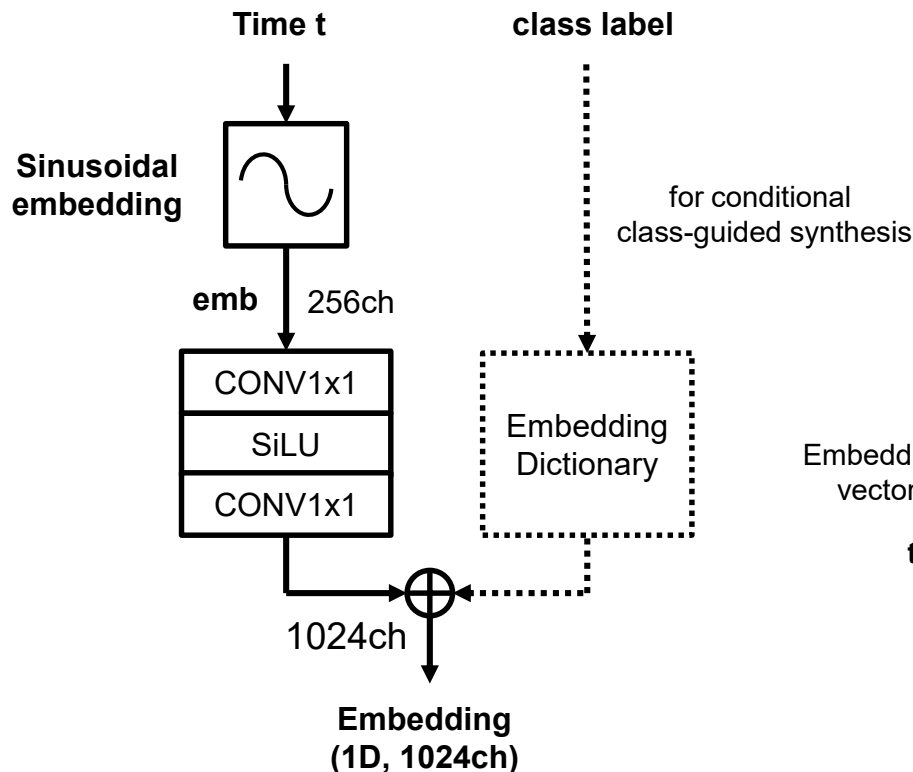
Embedding

(ADM for ImageNet 256)

Sinusoidal embedding (positional encoding):

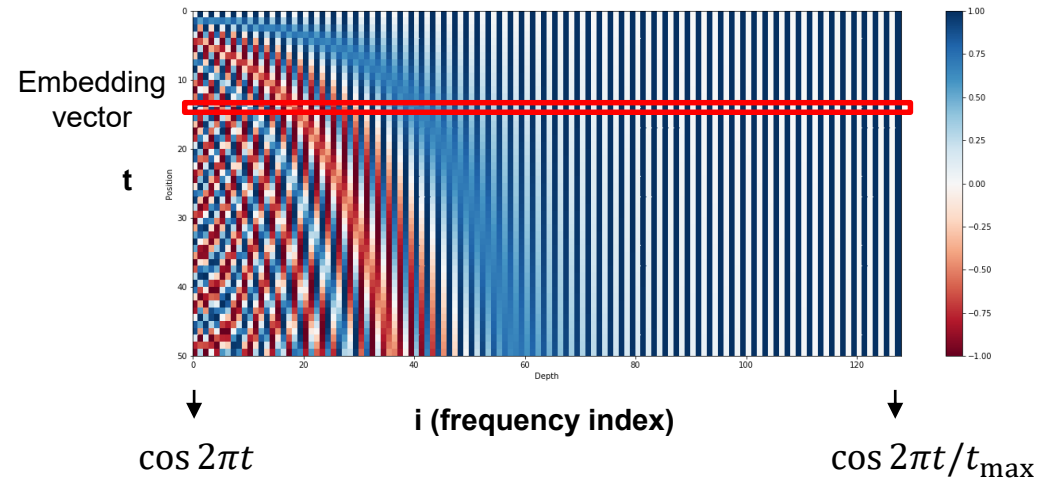
$$\text{emb}[i] = \begin{cases} \cos \omega_i t, & i \text{ is even} \\ \sin \omega_i t, & i \text{ is odd} \end{cases} \quad \omega_i = \frac{2\pi}{t_{\max} \lfloor \frac{i}{2} \rfloor / (\frac{C}{2})}$$

$$t_{\max} = 10000, C = 256$$



Example for C=128:

A row vector is an embedding for a specific t
(credit: https://kazemnejad.com/blog/transformer_architecture_positional_encoding/)



High-frequency features for learning the responses of small variations, facilitating small-step interpolation

One-to-one linear-to-circular mapping

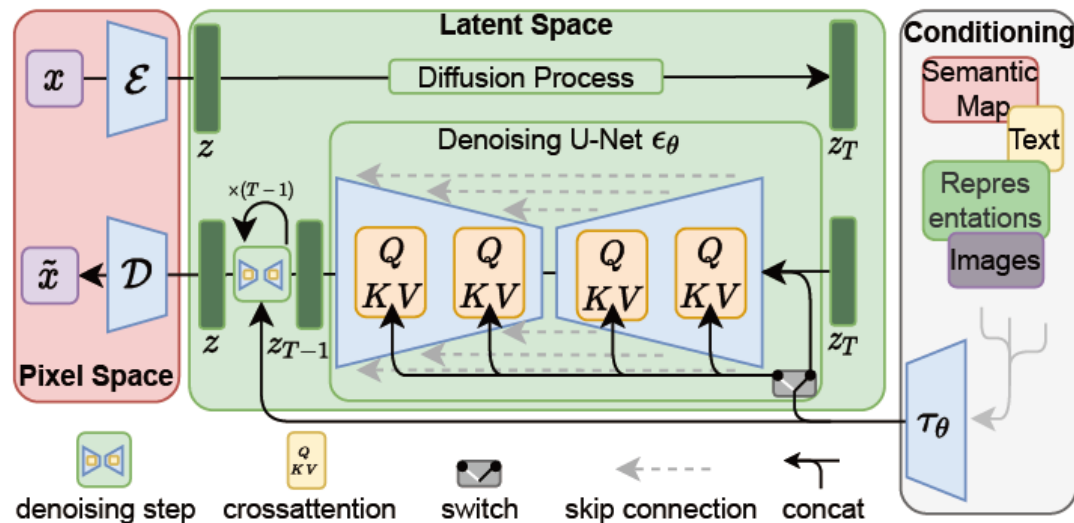


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Latent diffusion model (LDM)

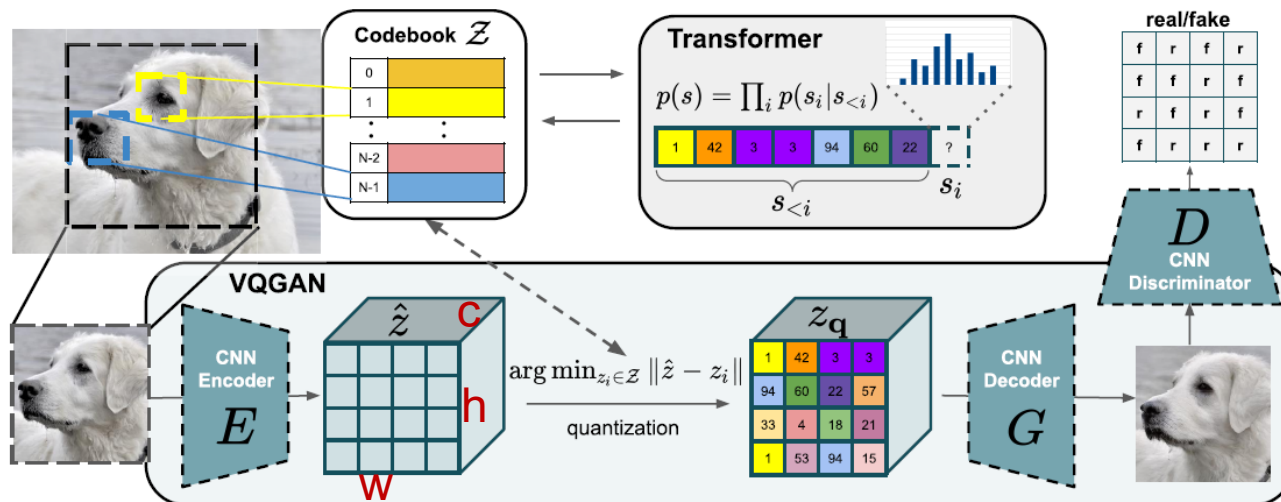
- Performs diffusion in latent space which is downsampled by 4-8x
- Proposes cross-attention for conditioning
 - Q from U-Net features
 - K and V from intermediate representation of the conditional input
- Becomes the foundation model for Stable Diffusion



Ref: Rombach et al., CVPR'22; LDM.

Perceptual compression via latent space

- VQ-regularized LDM adopts VQGAN encoder/decoder
 - CNN encoder: $H \times W \times 3 \rightarrow h \times w \times c$
 - Downsample factor $f = \frac{H}{h} = \frac{W}{w}$
 - Vector quantization (VQ): c -tuple vector $\rightarrow$ entry of codebook $\mathcal{Z}$
 - CNN decoder (+single-head attention): $h \times w \times c \rightarrow H \times W \times 3$
 - LDM decoder = VQ+CNN decoder



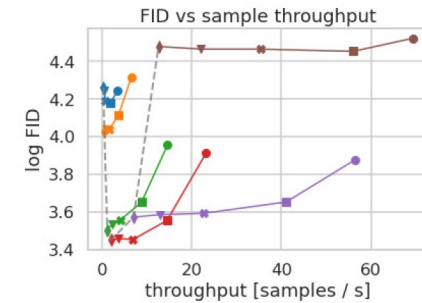
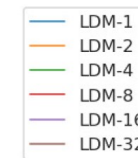
Ref: Esser et al., CVPR'21; VQGAN.



Experimental results

Models for ImageNet

	<i>LDM-1</i>	<i>LDM-2</i>	<i>LDM-4</i>	<i>LDM-8</i>
<i>z</i> -shape	$256 \times 256 \times 3$	$128 \times 128 \times 2$	$64 \times 64 \times 3$	$32 \times 32 \times 4$
$ \mathcal{Z} $	-	2048	8192	16384
Diffusion steps	1000	1000	1000	1000
Noise Schedule	linear	linear	linear	linear
Model Size	396M	391M	391M	395M
Channels	192	192	192	256
Depth	2	2	2	2
Channel Multiplier	1,1,2,2,4,4	1,2,2,4,4	1,2,3,5	1,2,4



Class-conditional ImageNet synthesis

Method	FID↓	IS↑	Precision↑	Recall↑	N_{params}	
BigGan-deep [3]	6.95	203.6 ± 2.6	0.87	0.28	340M	-
ADM [15]	10.94	100.98	0.69	0.63	554M	250 DDIM steps
ADM-G [15]	<u>4.59</u>	186.7	<u>0.82</u>	0.52	608M	250 DDIM steps
<i>LDM-4</i> (ours)	10.56	103.49 ± 1.24	0.71	<u>0.62</u>	400M	250 DDIM steps
<i>LDM-4-G</i> (ours)	3.60	247.67 ± 5.59	0.87	0.48	400M	250 steps, c.f.g [32], $s = 1.5$

Efficiency for inpainting

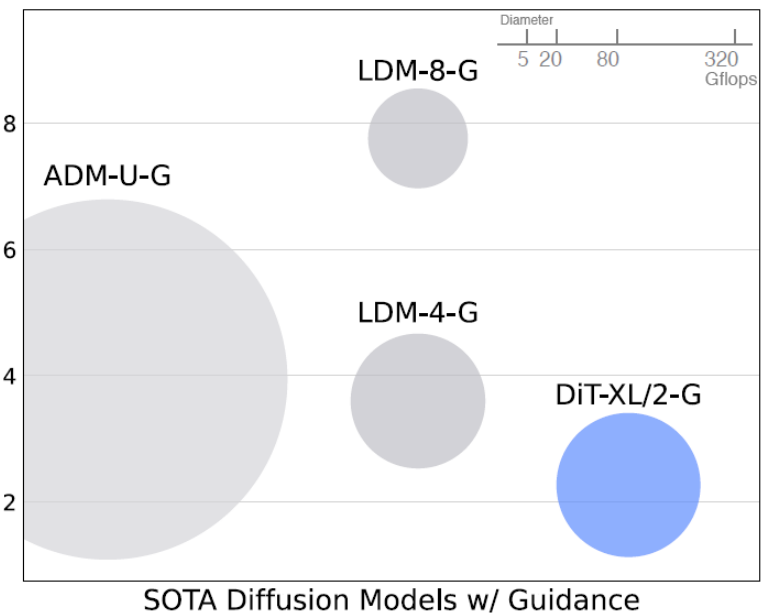
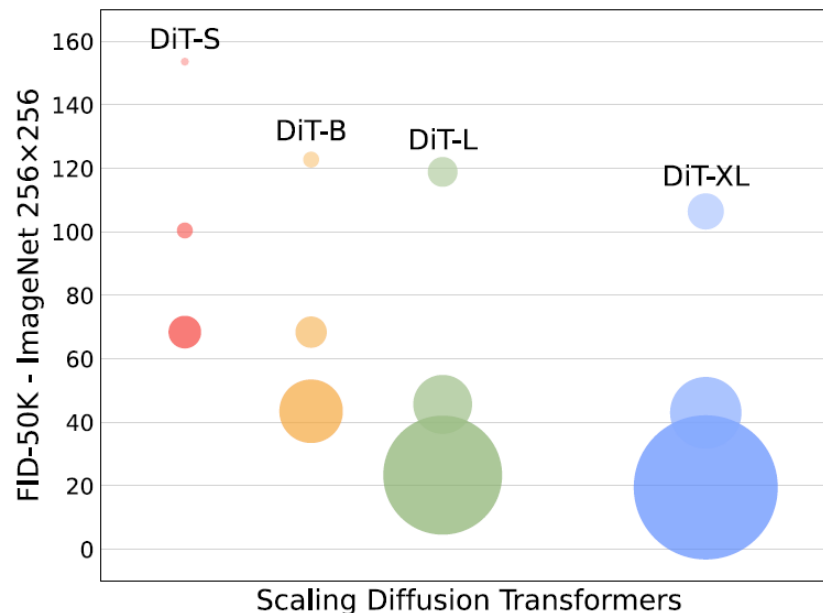
Model (<i>reg.</i> -type)	train throughput samples/sec.	sampling throughput [†] @256	sampling throughput [†] @512	train+val hours/epoch	FID@2k epoch 6
<i>LDM-1</i> (no first stage)	0.11	0.26	0.07	20.66	24.74
<i>LDM-4</i> (<i>KL</i> , w/ attn)	0.32	0.97	0.34	7.66	15.21
<i>LDM-4</i> (<i>VQ</i> , w/ attn)	0.33	0.97	0.34	7.04	14.99
<i>LDM-4</i> (<i>VQ</i> , w/o attn)	0.35	0.99	0.36	6.66	15.95

5x



Diffusion transformer (DiT) [ICCV'23]

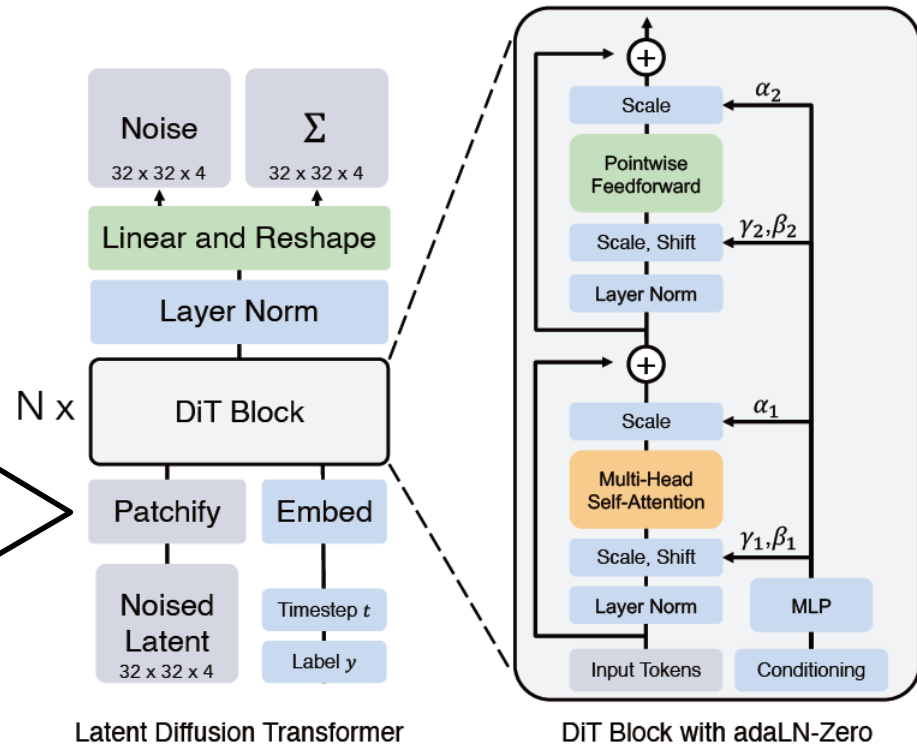
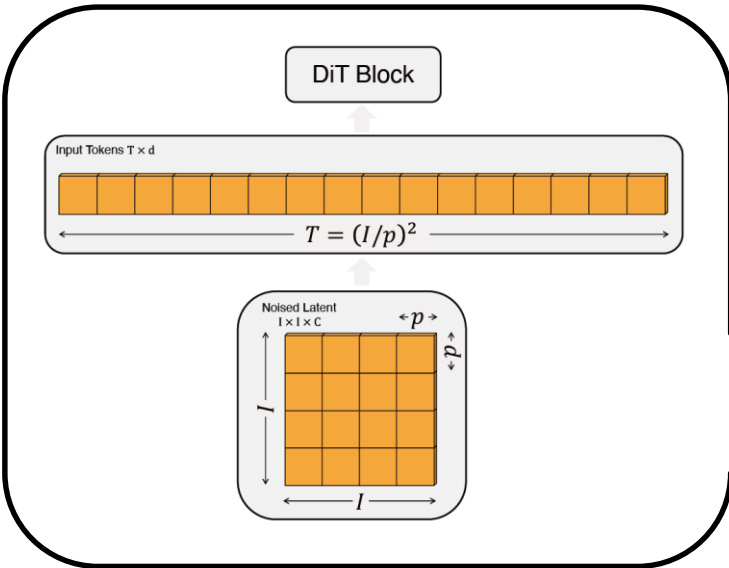
- Explore the scalability of pure transformer-based DMs
 - Diffuse in latent space as LDM
- Show the competitiveness of DiT
 - Compute-efficient compared to U-Net-based DMs
 - U-Net is not necessary for DM



Ref: Peebles et al., ICCV'23.

DiT architecture

- Cascades DiT blocks without upsampling and downsampling
- Applies adaptive layer norm with zero initialization (adaLN-zero) for conditioning
- Patchifies $I \times I \times C$ input to T tokens of d -channel features
 - Patch: $p \times p \times C \rightarrow d$

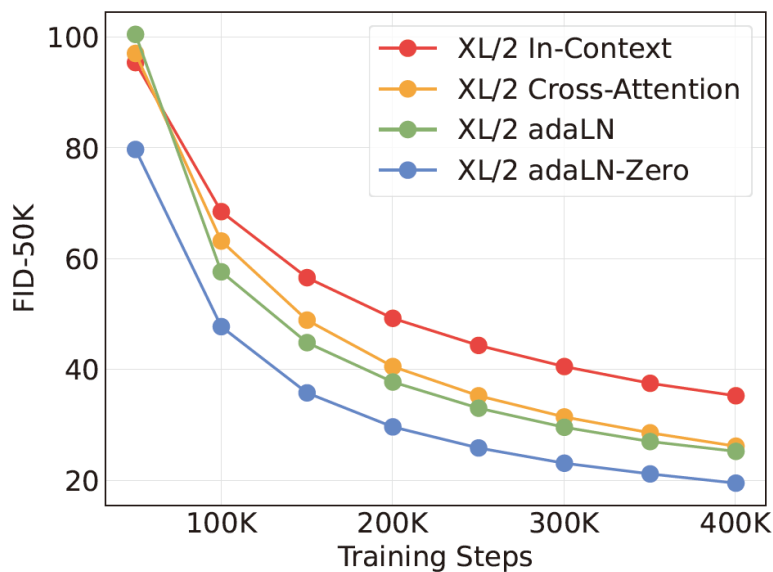




Experimental results

Model	Layers N	Hidden size d	Heads
DiT-S	12	384	6
DiT-B	12	768	12
DiT-L	24	1024	16
DiT-XL	28	1152	16

Comparison of conditioning strategies ($p=2$)

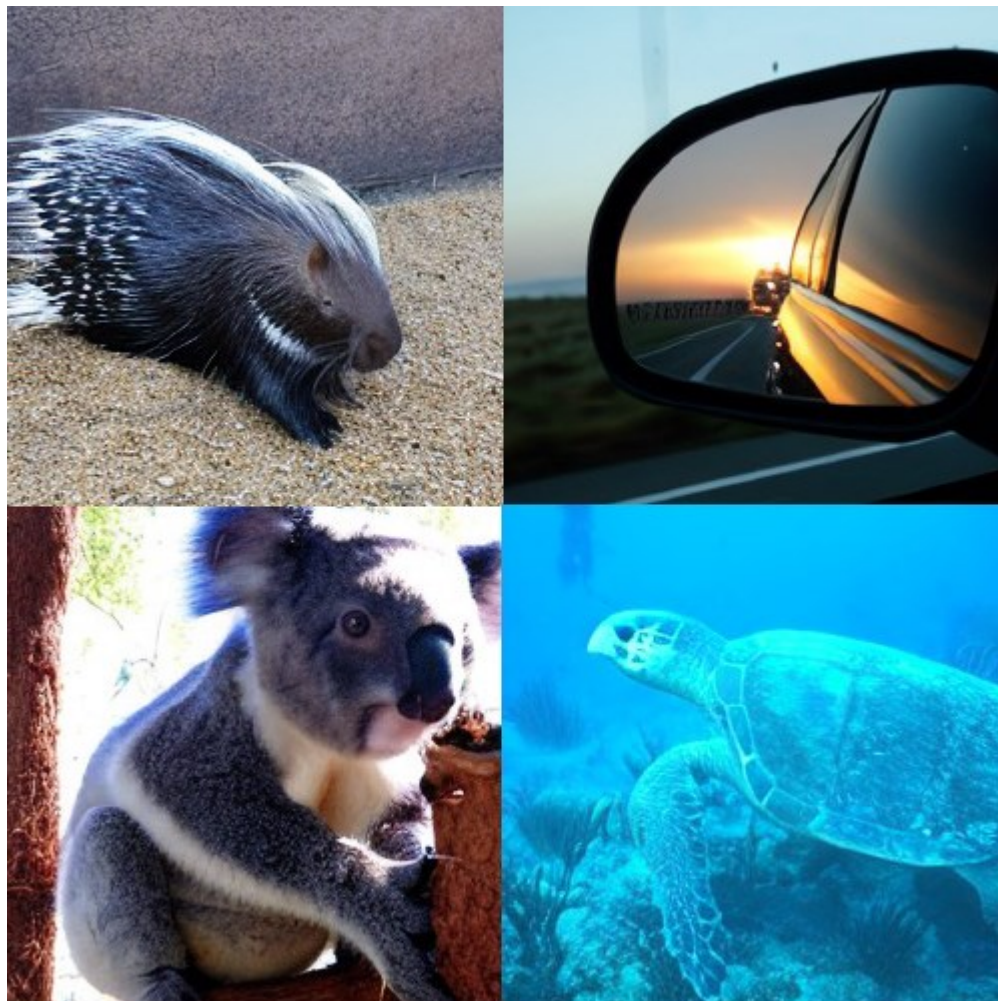
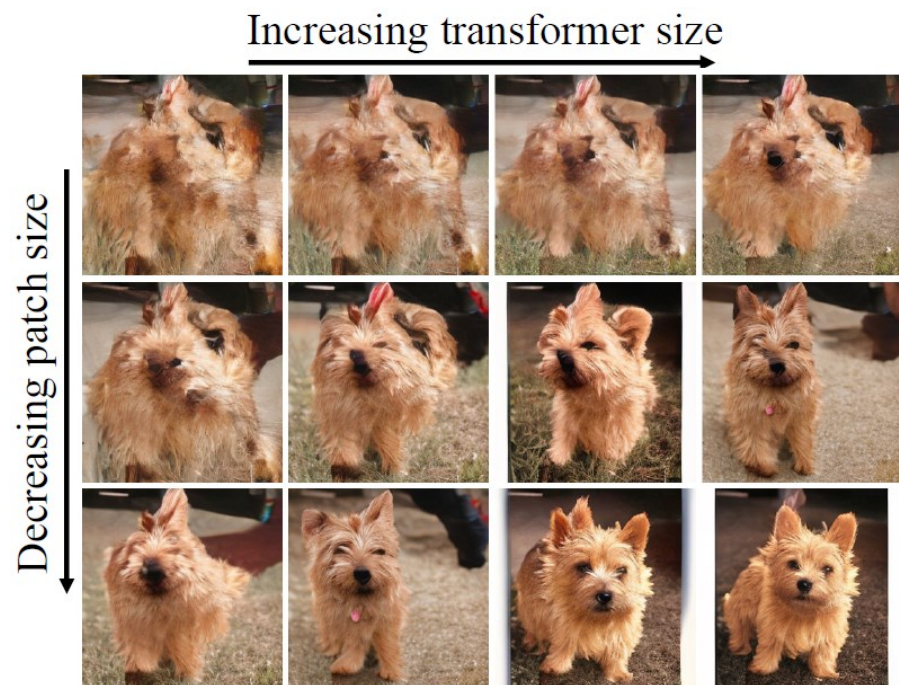


Class-Conditional ImageNet 256×256

Model	FID↓	sFID↓	IS↑	Precision↑	Recall↑
BigGAN-deep [2]	6.95	7.36	171.4	0.87	0.28
StyleGAN-XL [53]	2.30	4.02	265.12	0.78	0.53
ADM [9]	10.94	6.02	100.98	0.69	0.63
ADM-U	7.49	5.13	127.49	0.72	0.63
ADM-G	4.59	5.25	186.70	0.82	0.52
ADM-G, ADM-U	3.94	6.14	215.84	0.83	0.53
CDM [20]	4.88	-	158.71	-	-
LDM-8 [48]	15.51	-	79.03	0.65	0.63
LDM-8-G	7.76	-	209.52	0.84	0.35
LDM-4	10.56	-	103.49	0.71	0.62
LDM-4-G (cfg=1.25)	3.95	-	178.22	0.81	0.55
LDM-4-G (cfg=1.50)	3.60	-	247.67	0.87	0.48
DiT-XL/2	9.62	6.85	121.50	0.67	0.67
DiT-XL/2-G (cfg=1.25)	3.22	5.28	201.77	0.76	0.62
DiT-XL/2-G (cfg=1.50)	2.27	4.60	278.24	0.83	0.57

Visualization

DiT/XL/2 on ImageNet 256x256





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Classifier-free guidance

- Deployed in large-scale DMs, including DALL-E 2, Stable Diffusion, and Imagen
- Generates the guidance directly, instead of gradient evaluation for classifiers
 - Inference with weighted scores
 - w is adjustable after training, and $w = -1$ for unconditional synthesis

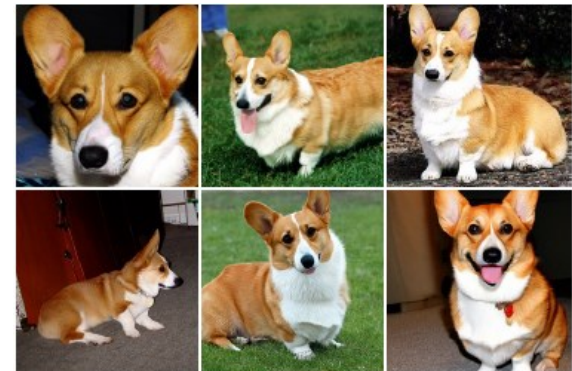
$$\tilde{\epsilon}_{\theta}(\mathbf{z}_{\lambda}, \mathbf{c}) = (1 + w)\epsilon_{\theta}(\mathbf{z}_{\lambda}, \mathbf{c}) - w\epsilon_{\theta}(\mathbf{z}_{\lambda})$$

- Trains the conditional and unconditional scores separately with a pre-defined selection probability
- Enables negative prompt (np)

$$\tilde{\epsilon}_{\theta}(z, c, np) = (1 + w)\epsilon_{\theta}(z, c) - w\epsilon_{\theta}(z, np)$$

- Note: CFG is essential for LDM.

$w = 3.0$

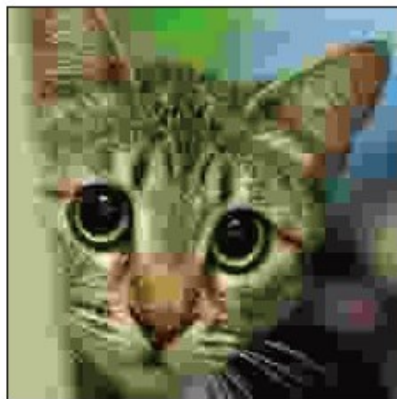


Ref: Ho et al., NeurIPS'21.

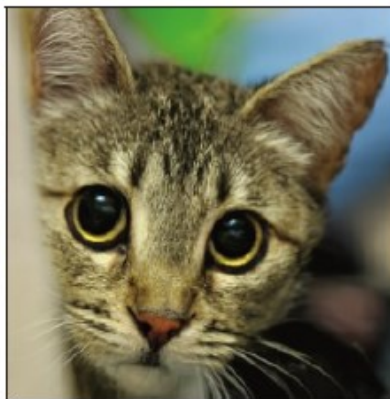
More examples for image-to-image translation (1/3): Palette

**JPEG
restoration
(JPEG-guided
synthesis)**

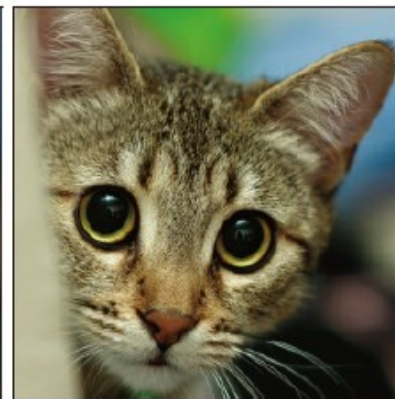
Input



Output



Original



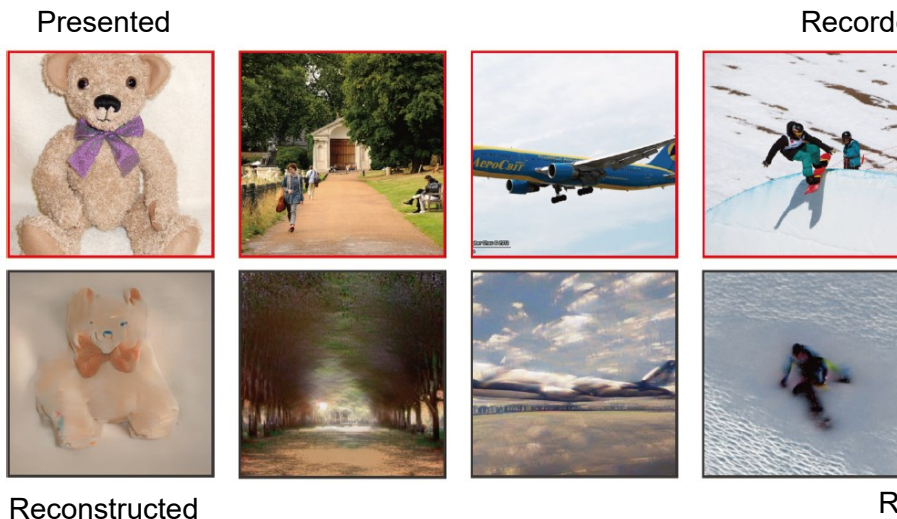
**Panorama
generation
(Uncropping)**



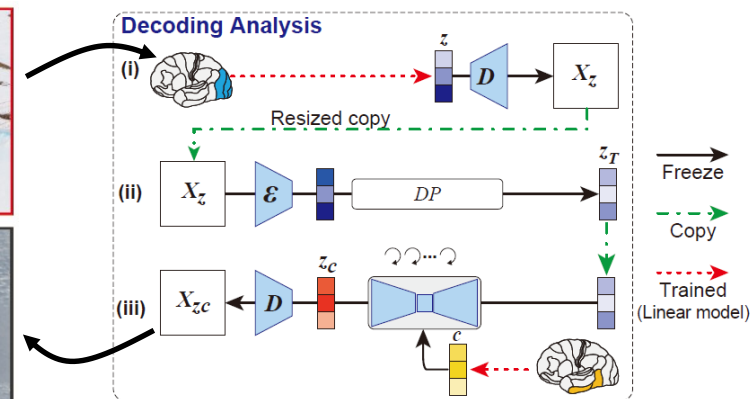
Ref: Saharia et al., SIGGRAPH'22.

More examples (2/3)

fMRI-to-Image translation



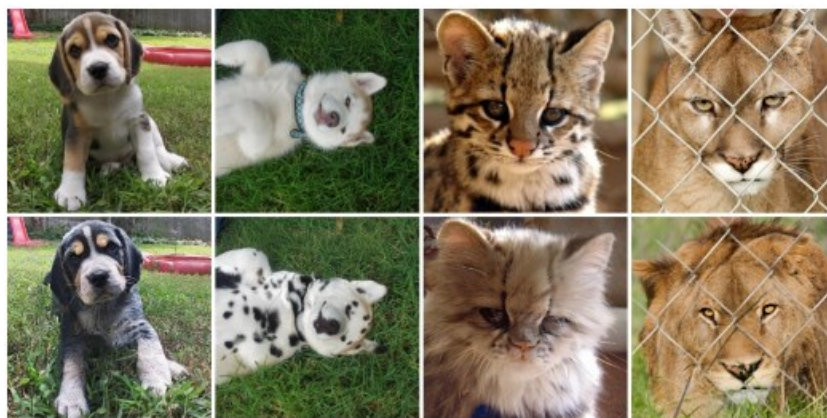
Recorded by fMRI



Reconstructed by embedding-guided DM

Ref: Takagi et al., CVPR'23.

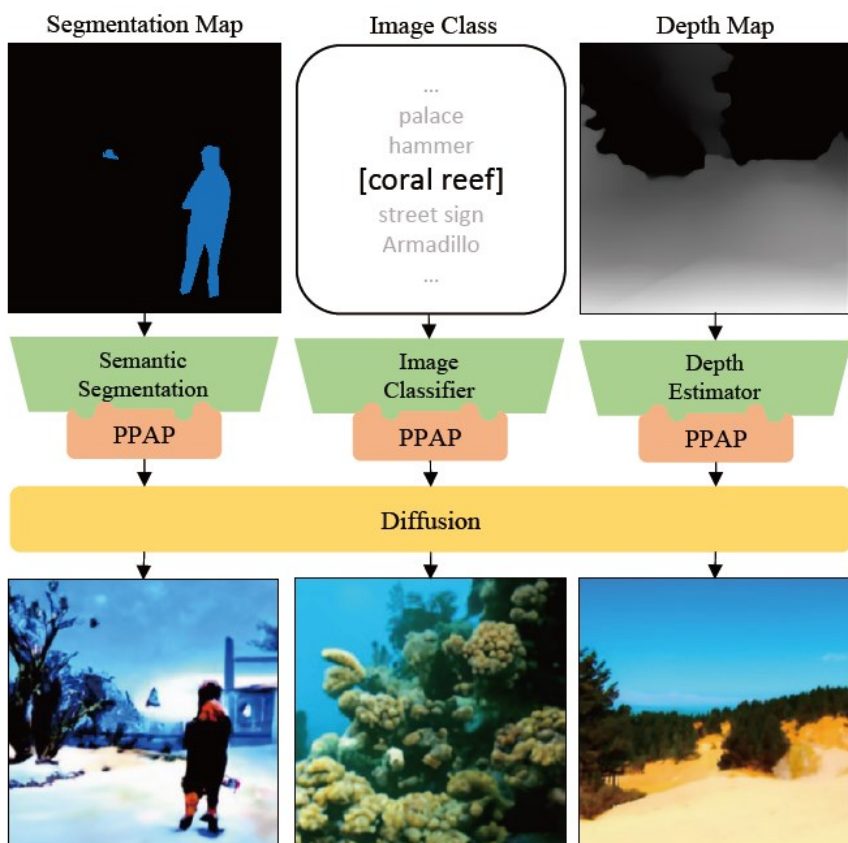
ImageNet-class Translation (DDIB)



Ref: Su et al., ICLR'23.

More examples for image-to-image translation (3/3)

Segmentation/class/depth to image
(PPAP: practical-plug-and-play)



Ref: Go et al., CVPR'23.

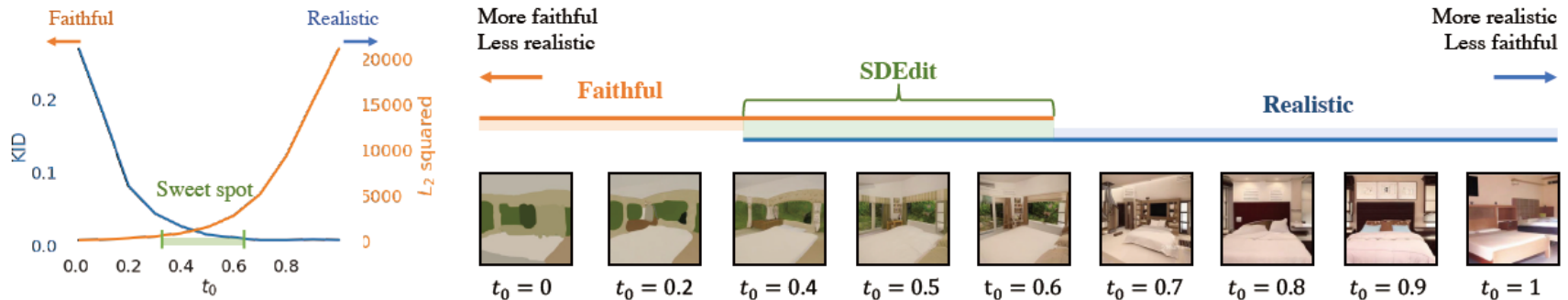
Label/edge to image
(BBDM: Brownian bridge DM)



Ref: Li et al., CVPR'23.

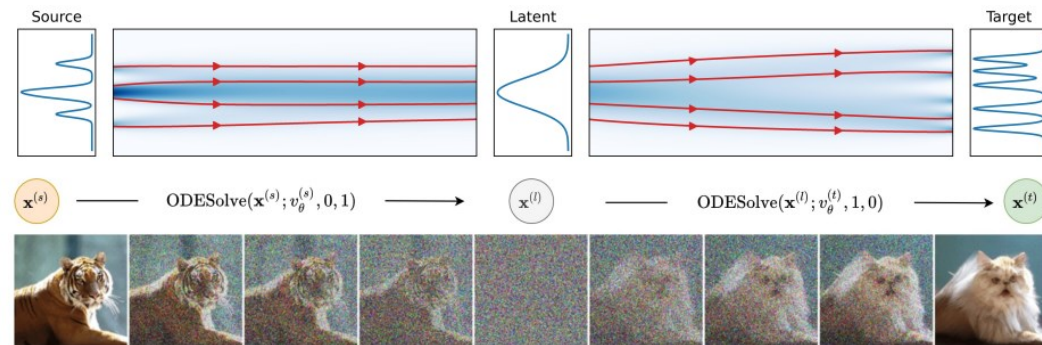
Approaches for image-to-image translation (1/3)

- Use off-the-shelf pre-trained DM
 - SDEdit [Meng et al., ICLR'22]: perturbs hand-edited images (e.g. stroke or composition) with certain noise σ_{t_0} and then reverses SDE



- Dual Diffusion Implicit Bridge (DDIB) [Su et al., ICLR'23]: converts source images into latent embeddings with domain-A DDIM and then reverses them with domain-B DDIM

VE SDEs are adopted to preserve skeletons in latent images.



Approaches for image-to-image translation (2/3)

- Train conditional DM
 - Palette [Saharia et al., SIGGRAPH'22]: image-conditional DM
 - Cascades input image as DM input
 - Verifies 1) L2 loss better than L1 on diversity; 2) self-attention is important
 - Practical Plug-and-play (PPAP) [Go et al., CVPR'23]: classifier-conditional

Palette

$$\mathbb{E}_{(x,y)} \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)} \mathbb{E}_{\gamma} \left\| f_{\theta}(x, \underbrace{\sqrt{\gamma} y + \sqrt{1-\gamma} \epsilon}_{\tilde{y}}, \gamma) - \epsilon \right\|_p^p$$

Input image

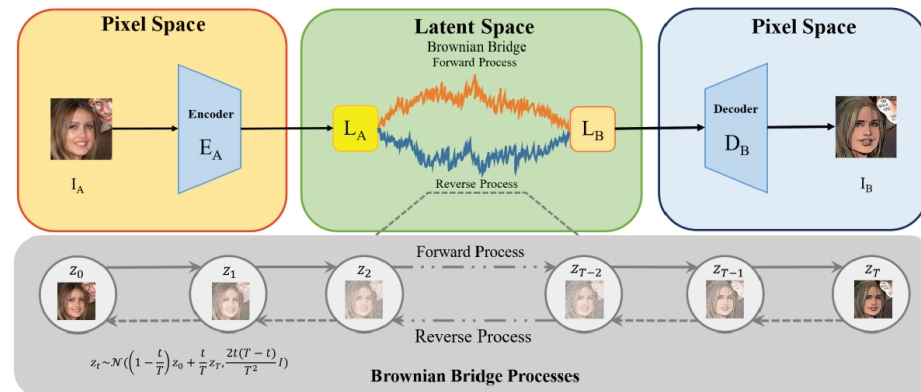
PPAP

$$x_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon_{\theta}(x_t, t) \right) + \sigma_t z - s \sigma_t \nabla_{x_t} \mathcal{L}_{\text{guide}}(f_{\phi}(x_t), y_{\text{target}})$$

Guiding loss Classifier-alike Input (segmentation/depth/class)

- Train paired-distribution transfer
 - Brownian Bridge Diffusion Model (BBDM) [Li et al., CVPR'23]: novel formulation with Brownian bridge

z_0 : source distribution
 z_T : target distribution



Approaches for image-to-image translation (3/3)

- Flow-based approach

- Rectified flow [Liu and Gong, ICLR'23]: sets up straight paths between two distributions (π_0, π_1) and learns the velocity field directly
 - No Gaussian assumption on distributions

$$\min_v \int_0^1 \mathbb{E} \left[\left\| (X_1 - X_0) - v(X_t, t) \right\|^2 \right] dt, \quad \text{with} \quad X_t = tX_1 + (1-t)X_0$$

$$X_0 \sim \pi_0, X_1 \sim \pi_1$$

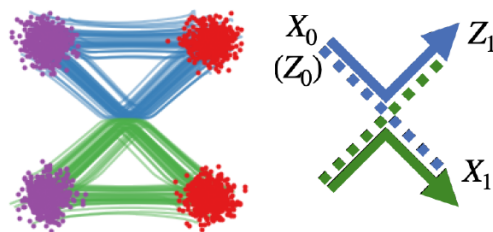
**Random coupling
in the beginning**



(a) Linear interpolation

$$X_t = tX_1 + (1-t)X_0$$

**Deterministic coupling
after learning velocity**



(b) Rectified flow Z_t

induced by (X_0, X_1)



(A) Cat $\rightarrow$ Wild Animals



(B) Wild Animals $\rightarrow$ Cat

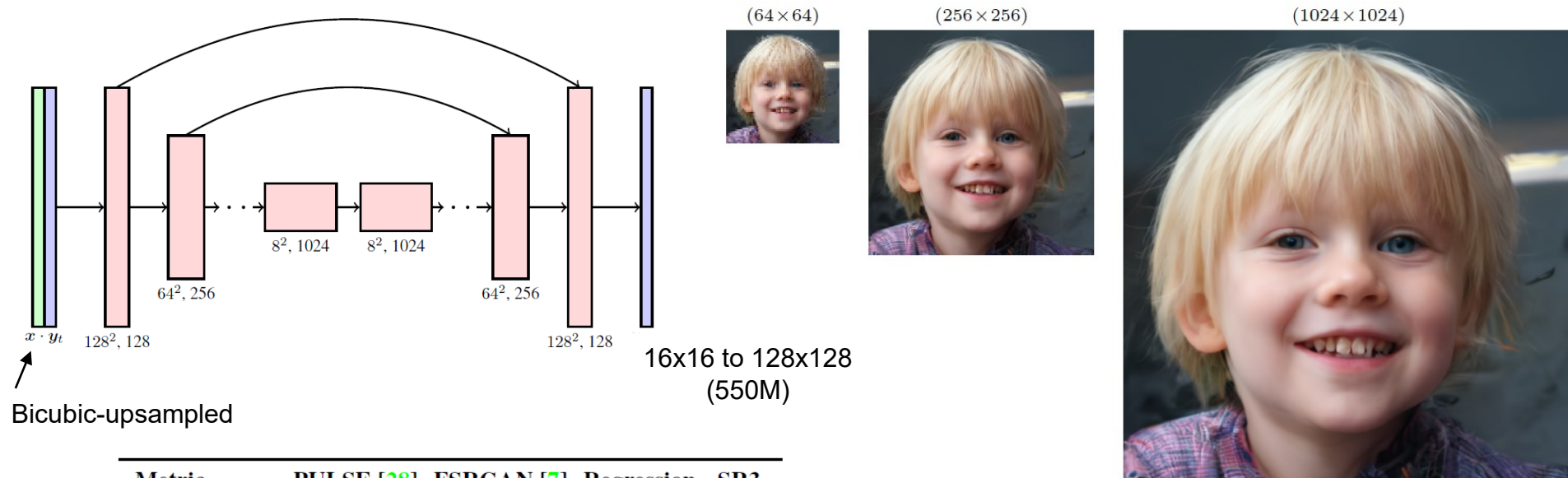


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SR3: Super-resolution via iterative refinement

- Embed bicubic-upsampled low-resolution image into U-Net
- Demonstrate the efficiency of cascaded diffusion models [JMLR'22]
 - E.g. 64x64 synthesis $\rightarrow$ SR to 256x256 $\rightarrow$ SR to 1024x1024



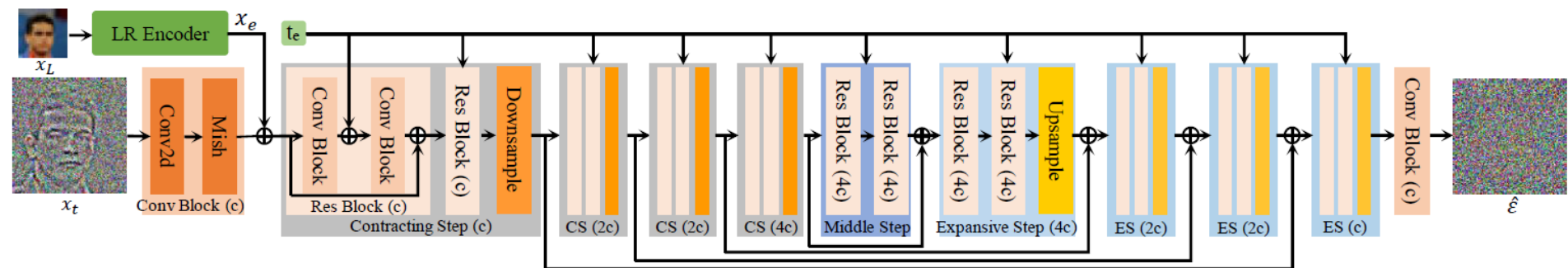
Metric	PULSE [28]	FSRGAN [7]	Regression	SR3
PSNR $\uparrow$	16.88	23.01	23.96	23.04
SSIM $\uparrow$	0.44	0.62	0.69	0.65
Consistency $\downarrow$	161.1	33.8	2.71	2.68

Consistency: MSE between LR and downsampled SR

Ref: Saharia et al., TPAMI'23; SR3.

SRDiff

- Embed the SR encoding of low-resolution image into U-Net
- Demonstrate the quality for lightweight and resolution-flexible models
 - No attention blocks to enable direct high-resolution synthesis
 - DM size: 12M (L1 loss, instead of L2); diffusion step: 100



DIV2K 0809/0815

LR

EDSR

RRDB

ESRGAN

RankSRGAN

SRFlow

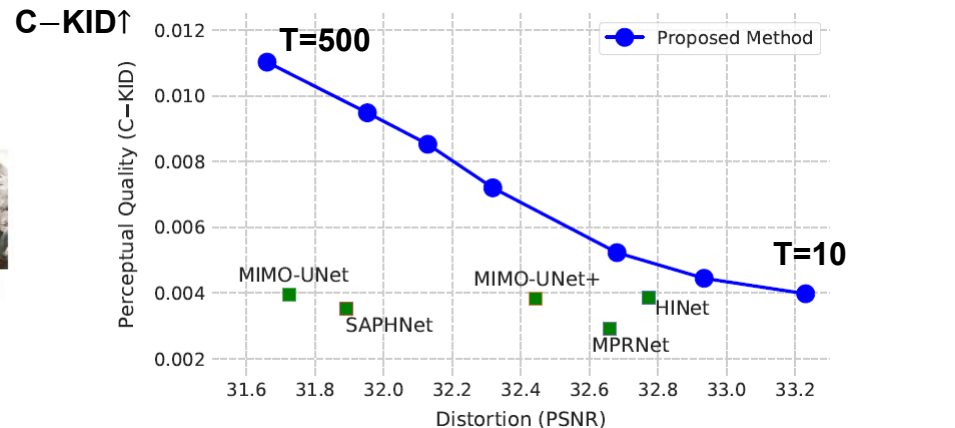
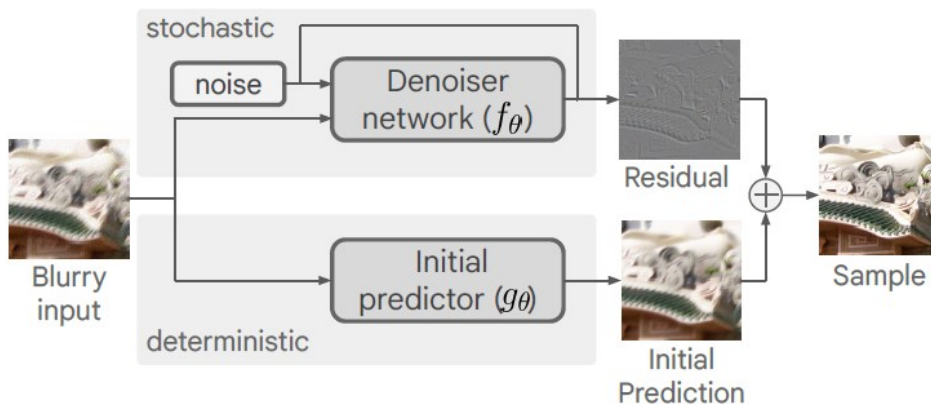
SRDiff

GT

Ref: Li et al., Neurocomputing 2022; SRDiff.

Deblurring via stochastic refinement

- Combine a deterministic predictor g_θ for one-time prediction and a stochastic DM denoiser f_θ for T-step residual refinement
- Demonstrate the quality for lightweight and resolution-flexible models
 - No attention blocks to enable region-agnostic restoration
 - DM size: 26M for g_θ and 7M for f_θ ; residues are easier to learn



$$\mathbb{E} \left\| \epsilon - f_\theta \left(\sqrt{\bar{\alpha}} \left(\underbrace{x_0 - g_\theta(y)}_{\text{residual}} \right) + \sqrt{1 - \bar{\alpha}} \epsilon, \bar{\alpha}, y \right) \right\|_1$$



Ref: Whang et al., CVPR'22.

ResShift (1/2)

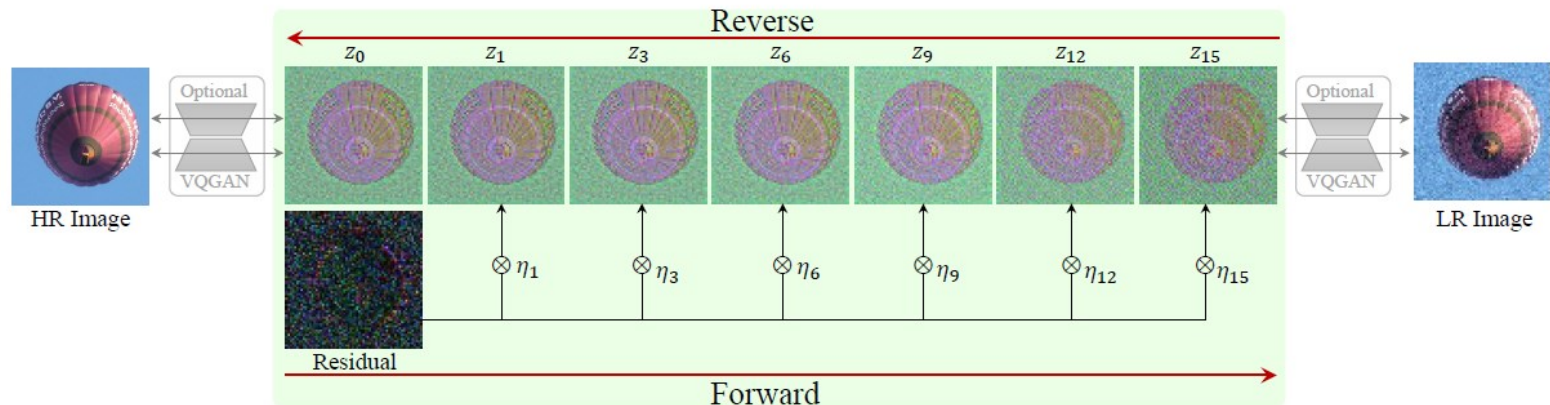
- Propose a new DPM scheme for super-resolution
 - Forward process: non-Markovian; HR ($\mathbf{x}_0$) to LR ($\mathbf{y}_0$)+noise

$$q(\mathbf{x}_t | \mathbf{x}_0, \mathbf{y}_0) = \mathcal{N}(\mathbf{x}_t; \underbrace{\mathbf{x}_0 + \eta_t \mathbf{e}_0}_{\eta_t \mathbf{y}_0 + (1 - \eta_t) \mathbf{x}_0}, \kappa^2 \eta_t \mathbf{I}), \quad t = 1, 2, \dots, T \quad \mathbf{e}_0 = \mathbf{y}_0 - \mathbf{x}_0 \quad \text{residual}$$

$$\eta_1 \rightarrow 0 \text{ and } \eta_T \rightarrow 1$$

- Reverse process: learnable p_θ

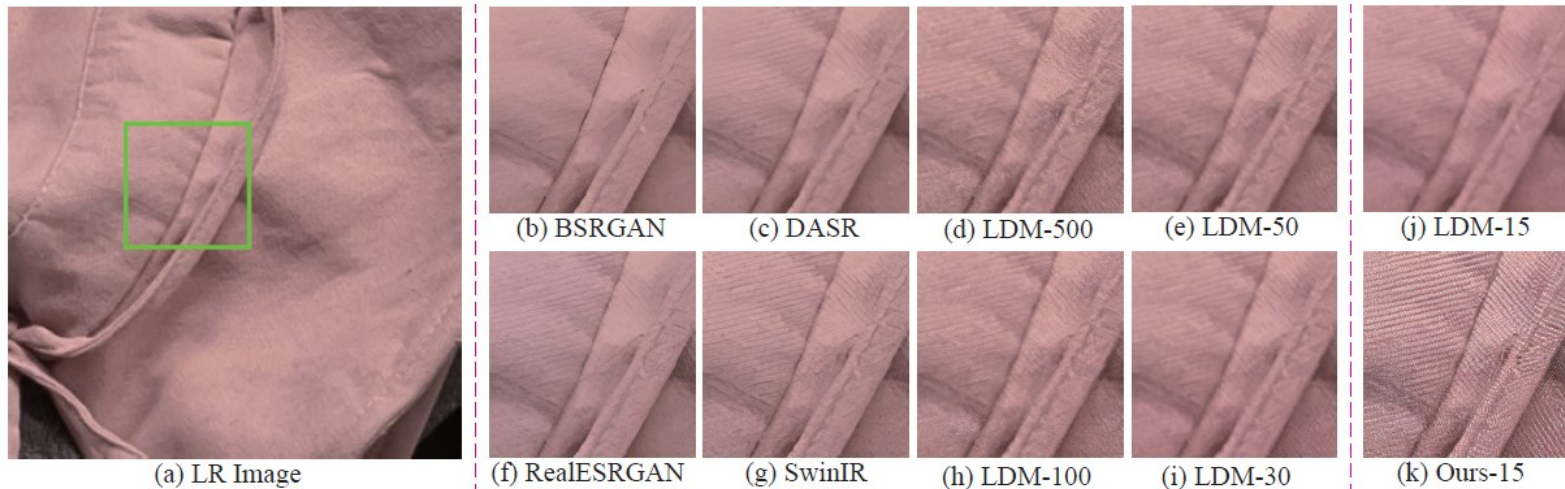
$$\min_{\theta} \sum_t D_{\text{KL}} [q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0, \mathbf{y}_0) || p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{y}_0)]$$



Ref: Yue et al., NIPS'23; ResShift.

ResShift (2/2)

- Enable 15-step high-quality inference
 - However, κ is set as small as 2.0



Metrics	Methods						
	BSRGAN	RealESRGAN	SwinIR	LDM-15	LDM-30	LDM-100	<i>ResShift</i>
PSNR $\uparrow$	24.42	24.04	23.99	24.89	24.49	23.90	25.01
LPIPS $\downarrow$	0.259	0.254	0.238	0.269	0.248	0.244	0.231
CLIPQA $\uparrow$	0.581	0.523	0.564	0.512	0.572	0.620	0.592
Runtime (s)	0.012	0.013	0.046	0.102	0.184	0.413	0.105
# Parameters (M)	16.70	16.70	28.01	113.60			118.59

Ref: Yue et al., NIPS'23; ResShift.



Outline

- **Implementation**
 - Image synthesis
 - Task-specific image restoration and enhancement
 - **Zero-shot image restoration (DM as pre-trained prior)**
 - Iterative Latent Variable Refinement (ILVR)
 - Denoising Diffusion Restoration Model (DDRM)
 - Denoising Diffusion Null-space Model (DDNM)
 - Generative Diffusion Prior (GDP)
 - Size-agnostic image generation
 - Video generation
 - Diffusion acceleration



Difficulty of applying DM as a general prior

- For guiding pre-trained SDE with distorted $\mathbf{y}_0$, we would like to use the corresponding score:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y}_0) = \underbrace{\nabla_{\mathbf{x}_t} \log p(\mathbf{y}_0 | \mathbf{x}_t)}_{\text{Observation model (likelihood) as data term}} + \underbrace{\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)}_{\text{Pre-trained DM as regularization}}$$

Observation model (likelihood) as data term Pre-trained DM as regularization

- But the distortion model is too complex to adopt:

Linear distortion: $\mathbf{y}_0 = \mathbf{K} \mathbf{x}_0 + \mathbf{z}$, $\mathbf{K}$ is the observation matrix and $\mathbf{z}$ is additive noise

Forward process: $\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}_t$, $\boldsymbol{\epsilon}_t \sim N(0, \mathbf{I})$

Observation model: $\mathbf{y}_0 = \mathbf{K} \frac{\mathbf{x}_t - \sigma_t \boldsymbol{\epsilon}_t}{\alpha_t} + \mathbf{z} \Rightarrow \mathbf{y}_0 \sim N\left(\frac{1}{\alpha_t} \mathbf{K} \mathbf{x}_t, \frac{\sigma_t^2}{\alpha_t^2} \mathbf{K} \mathbf{K}^T + \boldsymbol{\Sigma}_z\right)$

$$\Rightarrow \log p(\mathbf{y}_0 | \mathbf{x}_t) = -\frac{1}{2} \left(\mathbf{y}_0 - \frac{1}{\alpha_t} \mathbf{K} \mathbf{x}_t \right)^T \underbrace{\left(\frac{\sigma_t^2}{\alpha_t^2} \mathbf{K} \mathbf{K}^T + \boldsymbol{\Sigma}_z \right)^{-1}}_{\text{Difficult to evaluate! (time-dependent and SVD required)}} \left(\mathbf{y}_0 - \frac{1}{\alpha_t} \mathbf{K} \mathbf{x}_t \right)$$

Difficult to evaluate! (time-dependent and SVD required)

Iterative Latent Variable Refinement (ILVR)

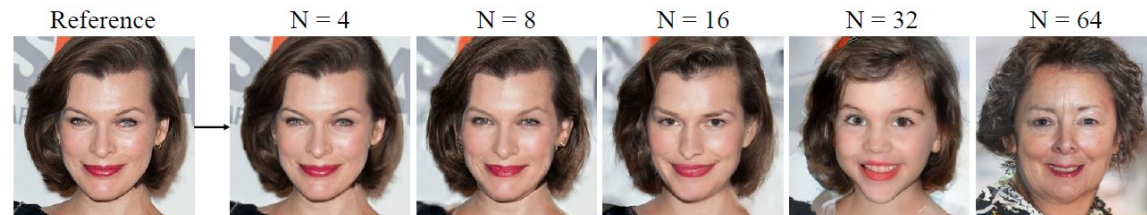
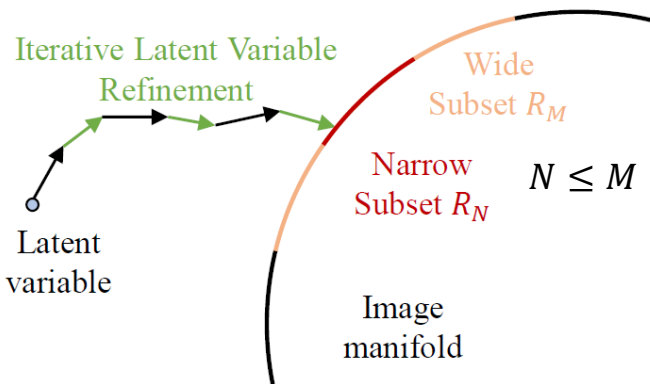
- Use low-passed input latent image $\phi_N(y_{t-1})$ to guide synthesis
 - Low-pass filter ϕ_N with scale N : downsample by N and then upsample back
 - N controls the range on image manifold
 - Direct extension for image restoration without explicit constraints

$$x'_{t-1} \sim p_{\theta}(x'_{t-1}|x_t) \quad \triangleright \text{unconditional proposal}$$

$$y_{t-1} \sim q(y_{t-1}|y) \quad \triangleright \text{condition encoding}$$

$$x_{t-1} \leftarrow x'_{t-1} + \phi_N(y_{t-1}) - \phi_N(x'_{t-1})$$

reference guide towards input image y



Ref: Choi et al., ICCV'21; ILVR.

Denoising Diffusion Restoration Model (DDRM)

$$\begin{aligned} y &= Hx + z \\ H &= U\Sigma V^T \end{aligned}$$

- For linear degradation H ,
 - Construct a diffusion process in spectral space $\bar{x} = V^T x$
 - Show that a pre-trained DM in spectral space with a reweighted objective equals to a pre-trained DM in pixel space x with additional spectral-space sampling

Spectral-space sampling (guidance) with $\bar{y} = \Sigma^\dagger U^T y$

$$\underbrace{x_t \rightarrow \epsilon_\theta(x_t) \rightarrow x_\theta(x_t)}_{\text{Pixel-space DM}} \rightarrow \underbrace{\bar{x}_\theta(x_t) = V^T x_\theta(x_t) \rightarrow \bar{x}_{t-1} \rightarrow x_{t-1} = V\bar{x}_{t-1}}_{\text{Spectral-space}}$$



Original



Low-res



DDRM (20)



DGP

(GAN-based)

Limitation

- Singular value decomposition (SVD) is required for H
- Complexity of evaluating $V^T x$ could be high, e.g. $O(|\dim(x)|^2)$
 - May be only practical for simple degradation, e.g. simple downsample kernel for SR

Ref: Kavar et al., NIPS'22; DDRM.



Denoising Diffusion Null-space Model (DDNM)

- For linear degradation A ,
 - Preserve the range-space part for data consistency
 - Update in the null space for realism
 - Work on data prediction ($x_\theta = x_{0|t}$)

$$y = Ax$$

$$x \equiv \underbrace{A^\dagger Ax}_{\text{Range space}} + \underbrace{(I - A^\dagger A)x}_{\text{Null space}} \quad (Ax_N = 0)$$

↓

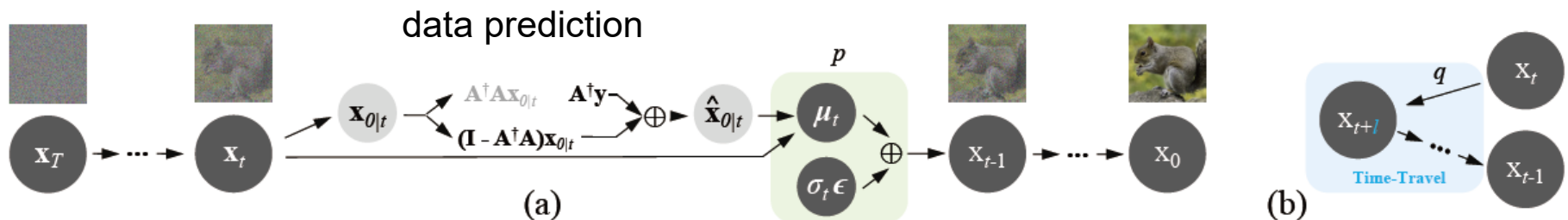
$$\hat{x}_{0|t} = A^\dagger y + (I - A^\dagger A)x_{0|t}$$

- DDNM⁺ further adds
 - Noise compensation with covariance matrices
 - Σ_t and Φ_t are used to confirm noise condition of $q(x_{t-1}|x_0)$
 - SVD of A may be required for accurate compensation
 - Travel-back trick with more NFEs

$$y = Ax + n$$

$$\hat{x}_{0|t} = x_{0|t} - \Sigma_t A^\dagger (Ax_{0|t} - y),$$

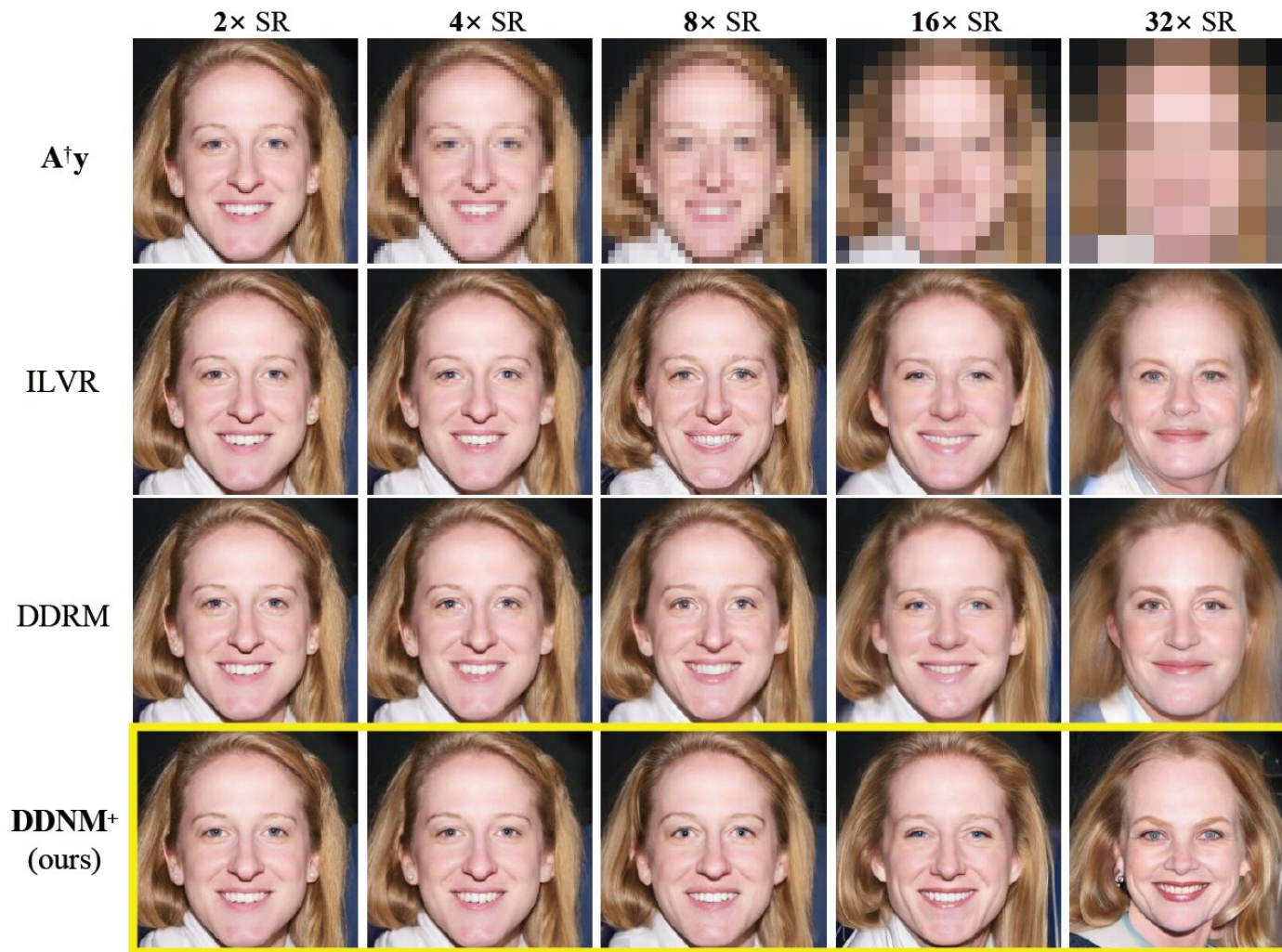
$$\hat{p}(x_{t-1}|x_t, \hat{x}_{0|t}) = \mathcal{N}(x_{t-1}; \mu_t(x_t, \hat{x}_{0|t}), \Phi_t I)$$



Ref: Wang et al., ICLR'23; DDNM.

Comparison for SR of CelebA

- DDNM outperforms in terms of PSNR, SSIM, and FID



Generative diffusion prior (GDP)

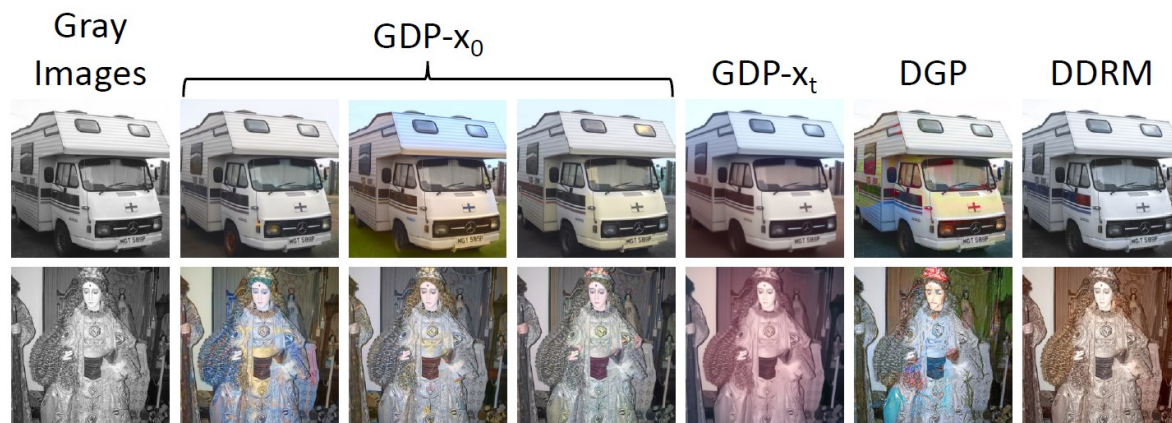
- Add heuristic gradient guidance to ImageNet-pretrained DM prior
 - Reconstruction loss $\mathcal{L}$ with degradation model $\mathcal{D}$ (data term with corrupted $\mathbf{y}$)
 - Quality enhancement loss $\mathcal{Q}$ (regularization term on target $\mathbf{x}$)
 - Better conditioned on estimated $\tilde{\mathbf{x}}_0$, instead of the current $\mathbf{x}_t$

$$\tilde{\mathbf{x}}_0 = \frac{\mathbf{x}_t}{\sqrt{\bar{\alpha}_t}} - \frac{\sqrt{1-\bar{\alpha}_t}\epsilon_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}}$$

$$\mathcal{L}_{\phi, \tilde{\mathbf{x}}_0}^{total} = \mathcal{L}(\mathbf{y}, \mathcal{D}_\phi(\tilde{\mathbf{x}}_0)) + \mathcal{Q}(\tilde{\mathbf{x}}_0)$$

Sample $\mathbf{x}_{t-1}$ by $\mathcal{N}(\mu + s\nabla_{\tilde{\mathbf{x}}_0}\mathcal{L}_{\phi, \tilde{\mathbf{x}}_0}^{total}, \Sigma)$

- Learn non-linear degradation for HDR recovery during inference



Ref: Fei et al., CVPR'23; GDP.



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 - **Size-agnostic image generation**
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Convolutional sampling for size extension

- Directly extending the trained parameters for larger resolution may be acceptable if the ratio is adequate
 - Self-/cross-attention trainable parameters are not directly related to resolution



Trained on 256^2 resolution but generalized to 512×1024
(semantic synthesis)

Ref: Rombach et al., CVPR'22; LDM.

Size-agnostic generation (1/3)

- DMs with self-attention impose size limitation, defying high-resolution image synthesis and image restoration
- Solutions for image synthesis:
 - Repeated SR DM, e.g. Imagen, for pre-defined image sizes
 - Uncropping DM, e.g. Palette, specifically trained for four-side image extension

256x2304 panorama extended from central 256x256 pixels

Input

Output



Size-agnostic generation (2/3)

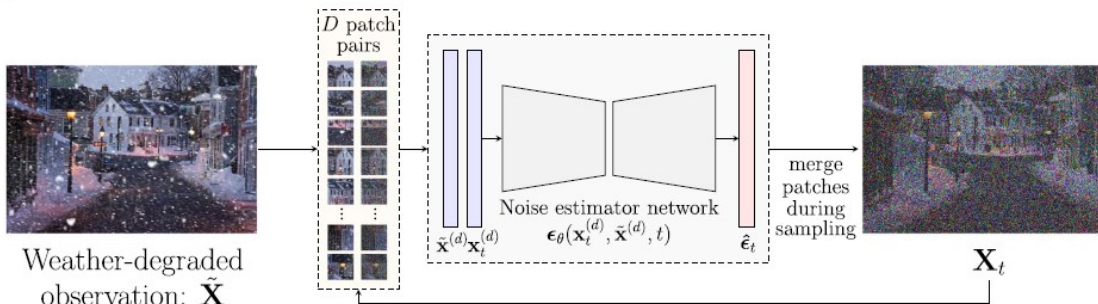
- Solutions for image restoration:
 - Fully convolutional network without self-attention, e.g. deblurring via stochastic refinement and SRDiff
 - Quality is acceptable for small-size DMs so far
 - Mask-shift trick in DDNM (null space) with complexity overheads
 - Generate a high-resolution image by gradually shifting masks
 - Data prediction for the masked region is replaced by the known clean pixels

$$\hat{\mathbf{x}}_{0|t} \text{ as } [\hat{\mathbf{x}}_{0|t}^{(left)}, \hat{\mathbf{x}}_{0|t}^{(right)}] \Rightarrow \hat{\mathbf{x}}_{0|t} = [\mathbf{x}^{(i)}, \hat{\mathbf{x}}_{0|t}^{(right)}]$$

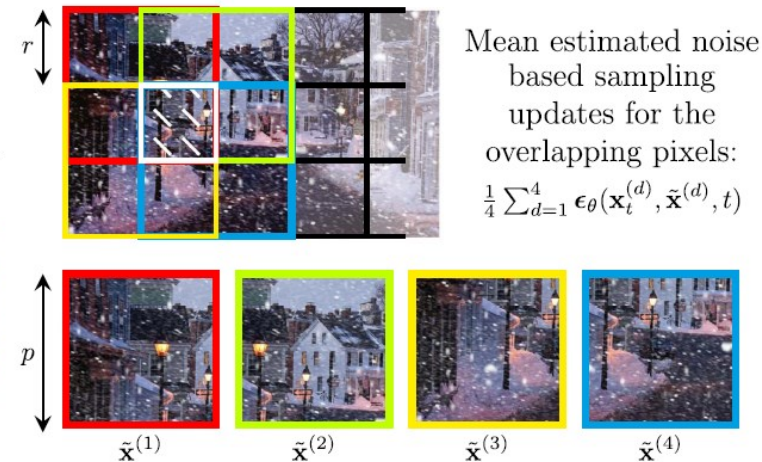


Size-agnostic generation (3/3)

- Patched-based DM [Özdenizci et al., TPAMI'23]
 - Train DM with cropped patches
 - Patch sizes of 64^2 and 128^2 are tested in the paper
 - Inference with overlapped patches with aggregated regions
 - Overlap size 16^2 has 4^2 and 8^2 times of complexity $\uparrow$, respectively
 - Averages the estimated noises from related patches



(a) Patch-based diffusive image restoration



(b) Illustrating sampling for overlapping patches

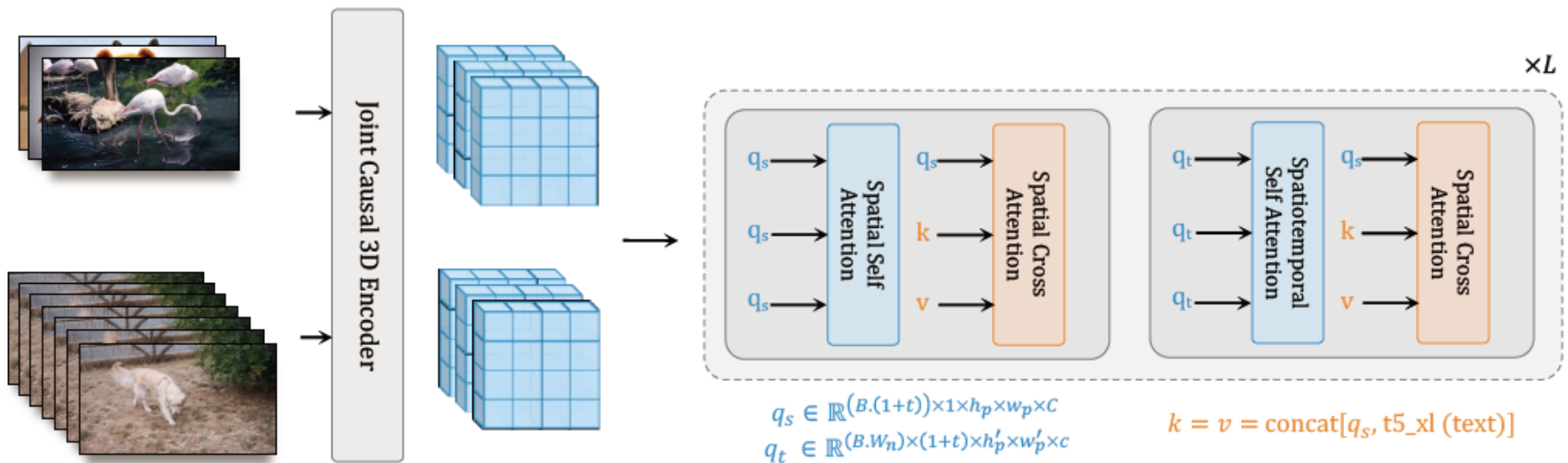


Outline

- **Implementation**
 - Image synthesis
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 - **Video generation**
 - **W.A.L.T.**
 - **Stable Video Diffusion**
 - **Sora**
 - Diffusion acceleration

Window Attention Latent Transformer (W.A.L.T.)

- Windowed attention on 3D latent space
 - 3D CNN encoder-decoder for video sequences
 - Transformer backbone
 - Spatial window: latent size $1 \times h_p \times w_p$
 - h_p, w_p : non-overlapped spatial patch size
 - Spatiotemporal window: latent size $(1 + t) \times h'_p \times w'_p$



Ref: Gupta et al., arXiv'23; W.A.L.T..

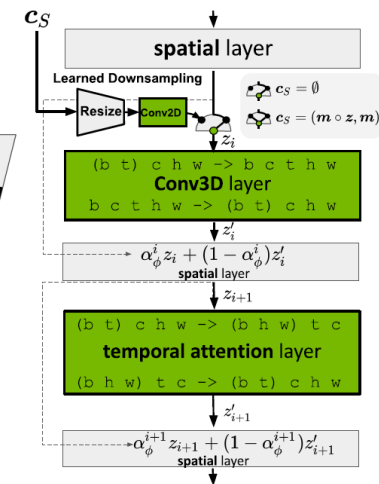
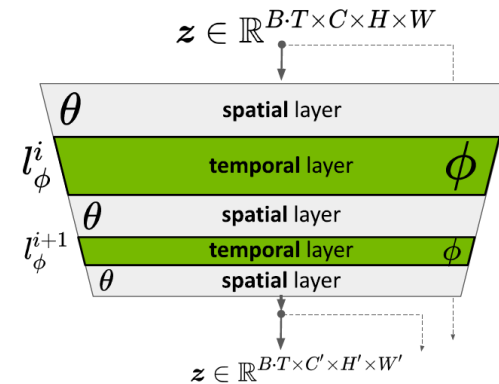
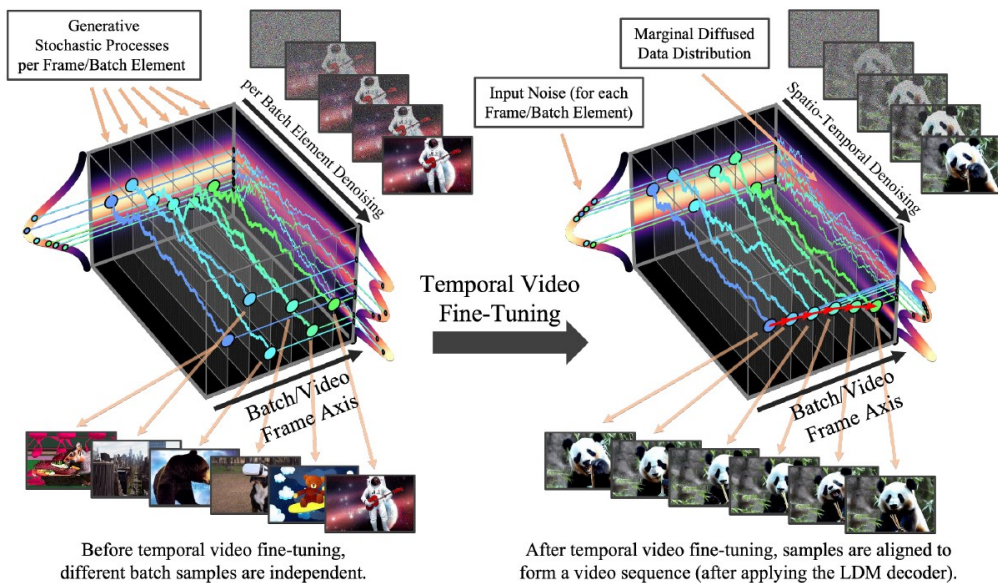
Result

- Base model + two video SR models to generate 24-frame 896x512 at 8fps
- “Cinematic action shot of a Ferrari driving on water”



Video LDM

- Add temporal layers to pre-trained image diffusion model
 - Base model generates key frames, interpolation models add intermediate frames, and SR model scales up resolution
 - Demonstrate 113-frame 2048x1280 at 24fps

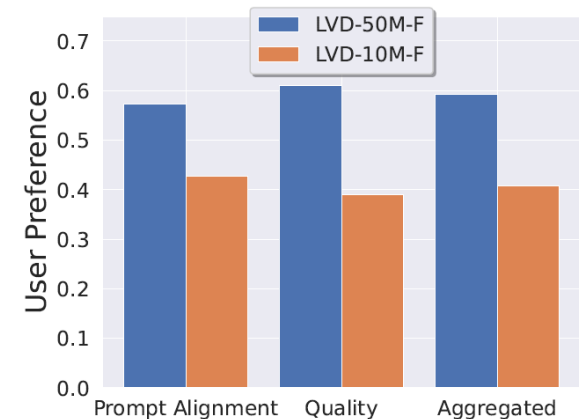
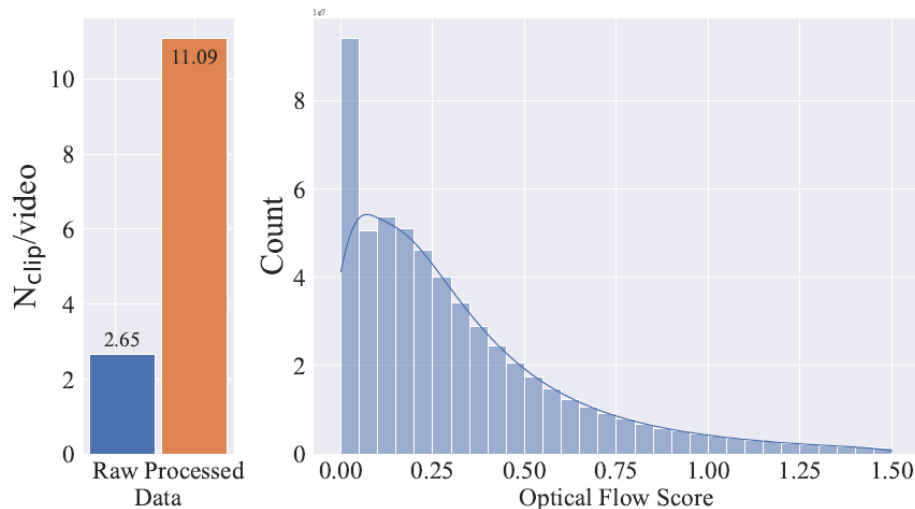


Ref: Blattmann et al., CVPR'23; Video LDM.



Stable Video Diffusion

- Enhances on base Video LDM with systematic video data curation
 - Image pre-training (SD2.1, 384x256)
 - video pre-training with curated data (14-frame 576x320)
 - video finetuning with small high-fidelity dataset (14-frame 1024x576)
 - finetune for 4x frame interpolation

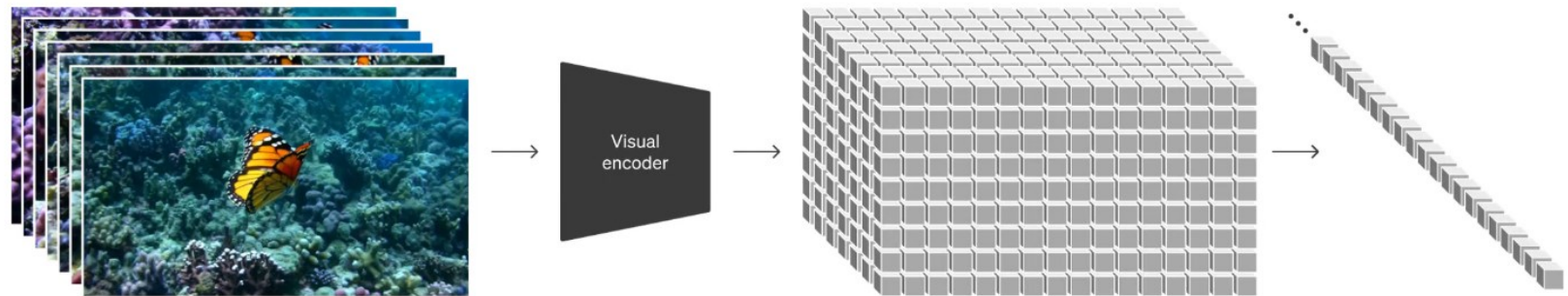


(d) User preference on scaling datasets.

Ref: Blattmann et al., Stability AI; SVD.

Sora

- Video generation models by Open AI
 - 3D latent space (space-time patches) + Diffusion transformer
 - Able to generate minute-long 1920x1080 30fps videos



Ref: <https://openai.com/sora>.

Result



- “An adorable kangaroo wearing blue jeans and a white t-shirt taking a pleasant stroll in Johannesburg South Africa during a colorful festival”



Animating DALL-E images



A Shiba Inu dog wearing a beret and black turtleneck.



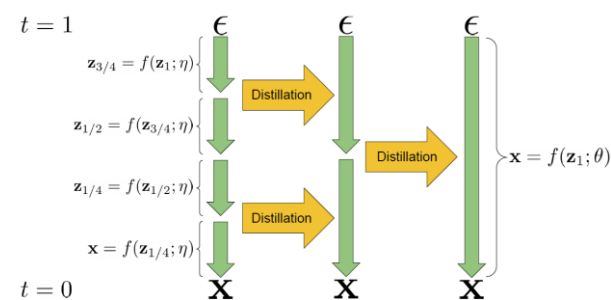
Outline

- **Implementation**
 - Image synthesis
 - Task-specific image restoration and enhancement
 - Zero-shot image restoration
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 - **Diffusion acceleration**
 - **Step reduction**
 - **Model distillation**
 - **Reflow**
 - Training-free samplers
 - Model shrinkage
 - Model quantization



Progressive distillation

- In each iteration, teach student model to learn 2-step **DDIM** prediction
- Through multiple iterations, have the model take as few as four steps without losing much perceptual quality



Formulation w/
data prediction $\hat{x}_\theta$

Teach the student model s.t.

$$z_{t''} = \alpha_{t''} \tilde{x} + \sigma_{t''} \frac{z_t - \alpha_t \tilde{x}}{\sigma_t}$$

Algorithm 1 Standard diffusion training

Require: Model $\hat{x}_\theta(z_t)$ to be trained
Require: Data set $\mathcal{D}$
Require: Loss weight function $w()$

while not converged **do**

$x \sim \mathcal{D}$ $\triangleright$ Sample data
 $t \sim U[0, 1]$ $\triangleright$ Sample time
 $\epsilon \sim N(0, I)$ $\triangleright$ Sample noise
 $z_t = \alpha_t x + \sigma_t \epsilon$ $\triangleright$ Add noise to data

$\tilde{x} = x$ $\triangleright$ Clean data is target for $\hat{x}$
 $\lambda_t = \log[\alpha_t^2 / \sigma_t^2]$ $\triangleright$ log-SNR
 $L_\theta = w(\lambda_t) \|\tilde{x} - \hat{x}_\theta(z_t)\|_2^2$ $\triangleright$ Loss
 $\theta \leftarrow \theta - \gamma \nabla_\theta L_\theta$ $\triangleright$ Optimization

end while

Algorithm 2 Progressive distillation

Require: Trained teacher model $\hat{x}_\eta(z_t)$
Require: Data set $\mathcal{D}$
Require: Loss weight function $w()$
Require: Student sampling steps N

for K iterations **do**

$\theta \leftarrow \eta$ $\triangleright$ Init student from teacher

while not converged **do**

$x \sim \mathcal{D}$
 $t = i/N, i \sim \text{Cat}[1, 2, \dots, N]$
 $\epsilon \sim N(0, I)$
 $z_t = \alpha_t x + \sigma_t \epsilon$
2 steps of DDIM with teacher
 $t' = t - 0.5/N, t'' = t - 1/N$
 $z_{t'} = \alpha_{t'} \hat{x}_\eta(z_t) + \frac{\sigma_{t'}}{\sigma_t} (z_t - \alpha_t \hat{x}_\eta(z_t))$
 $z_{t''} = \alpha_{t''} \hat{x}_\eta(z_{t'}) + \frac{\sigma_{t''}}{\sigma_{t'}} (z_{t'} - \alpha_{t'} \hat{x}_\eta(z_{t'}))$
 $\tilde{x} = \frac{z_{t''} - (\sigma_{t''}/\sigma_t) z_t}{\alpha_{t''} - (\sigma_{t''}/\sigma_t) \alpha_t}$ $\triangleright$ Teacher $\hat{x}$ target

$\lambda_t = \log[\alpha_t^2 / \sigma_t^2]$
 $L_\theta = w(\lambda_t) \|\tilde{x} - \hat{x}_\theta(z_t)\|_2^2$
 $\theta \leftarrow \theta - \gamma \nabla_\theta L_\theta$

end while

$\eta \leftarrow \theta$ $\triangleright$ Student becomes next teacher

$N \leftarrow N/2$ $\triangleright$ Halve number of sampling steps

end for

Ref: T. Salimans et al., ICLR'22.



Model parameterization and training loss

- Progressive distillation depends on training momentum of data prediction, and error prediction does not work well.
 - Model parameterization: conventional error-only prediction is not preferred; the connection between $\hat{x}_\theta$ and $\hat{\epsilon}_\theta$ breaks down in the case of the lowest SNR ratio ($\hat{\epsilon}_\theta(z_t) \rightarrow z_t$ as $t \rightarrow 1$) which infers nearly no information about data x_0 and occurs very frequently for small step numbers.
 - Training loss: gradients could be very small in that case.

$$L_\theta = \|\epsilon - \hat{\epsilon}_\theta(\mathbf{z}_t)\|_2^2 = \left\| \frac{1}{\sigma_t}(\mathbf{z}_t - \alpha_t \mathbf{x}) - \frac{1}{\sigma_t}(\mathbf{z}_t - \alpha_t \hat{\mathbf{x}}_\theta(\mathbf{z}_t)) \right\|_2^2 = \frac{\alpha_t^2}{\sigma_t^2} \|\mathbf{x} - \hat{\mathbf{x}}_\theta(\mathbf{z}_t)\|_2^2$$

Conventional/simple loss

- Alternatives of model parameterization (U-Net output)
 - Only $\hat{x}_\theta$
 - Combined $\hat{x}_\theta$ and $\hat{\epsilon}_\theta$
 - Novel **v-prediction** $\hat{v}_\theta := \alpha_t \hat{\epsilon}_\theta - \sigma_t \hat{x}_\theta$ (VP only) $\hat{\mathbf{x}} = \alpha_t \mathbf{z}_t - \sigma_t \hat{\mathbf{v}}_\theta(\mathbf{z}_t)$

Alternatives of training loss

- Truncated SNR weighting: $L_\theta = \max(\|\mathbf{x} - \hat{\mathbf{x}}_t\|_2^2, \|\epsilon - \hat{\epsilon}_t\|_2^2) = \max(\frac{\alpha_t^2}{\sigma_t^2}, 1) \|\mathbf{x} - \hat{\mathbf{x}}_t\|_2^2$
- SNR+1 weighting: $L_\theta = \|\mathbf{v}_t - \hat{\mathbf{v}}_t\|_2^2 = (1 + \frac{\alpha_t^2}{\sigma_t^2}) \|\mathbf{x} - \hat{\mathbf{x}}_t\|_2^2$



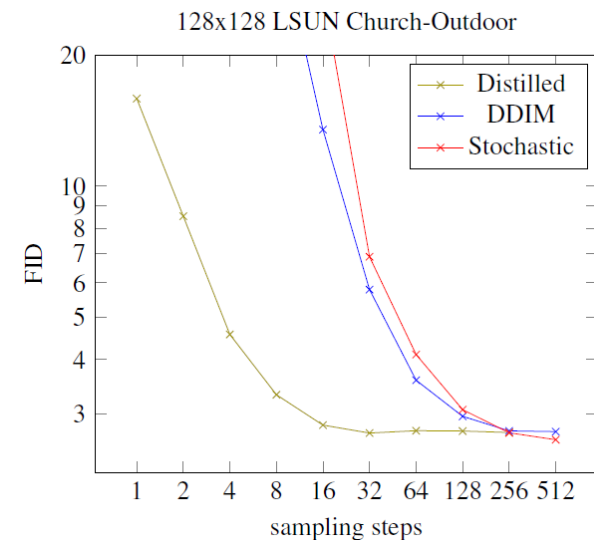
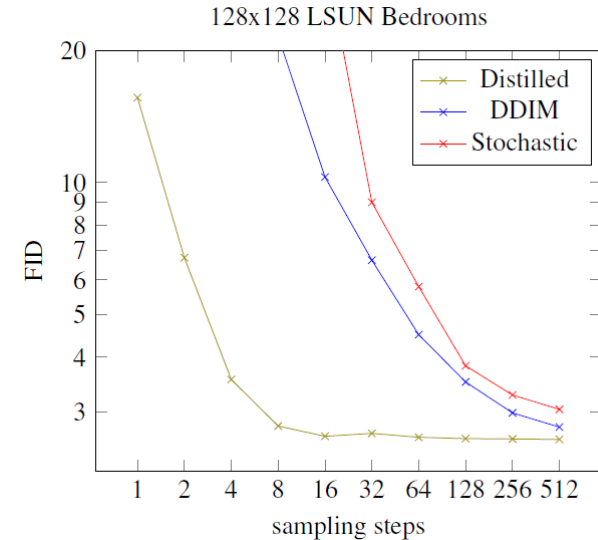
Experimental results

Distillation quality

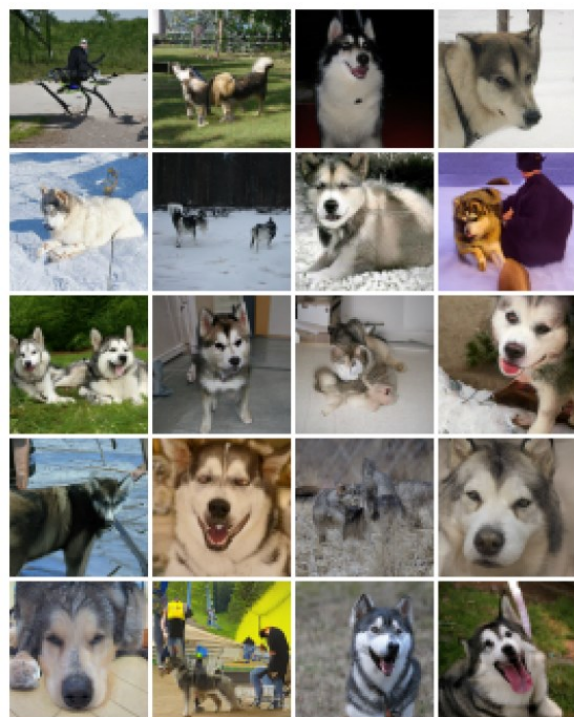
Unconditional CIFAR-10

Network Output	Loss Weighting	Stochastic sampler	DDIM sampler
$(\mathbf{x}, \epsilon)$ combined	SNR	2.54/9.88	2.78/9.56
	Truncated SNR	2.47/9.85	2.76/9.49
	SNR+1	2.52/9.79	2.87/9.45
$\mathbf{x}$	SNR	2.65/9.80	2.75/9.56
	Truncated SNR	2.53/ 9.92	2.51/9.58
	SNR+1	2.56/9.84	2.65/9.52
ϵ	SNR	2.59/9.84	2.91/9.52
	Truncated SNR	N/A	N/A
	SNR+1	2.56/9.77	3.27/9.41
$\mathbf{v}$	SNR	2.65/9.86	3.05/9.56
	Truncated SNR	2.45/9.80	2.75/9.52
	SNR+1	2.49/9.77	2.87/9.43

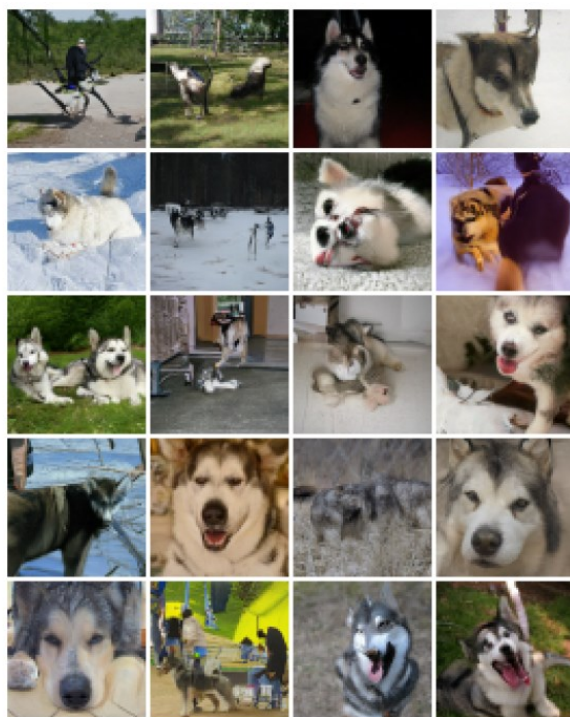
$\mathbf{v}$ -prediction shows robustness.



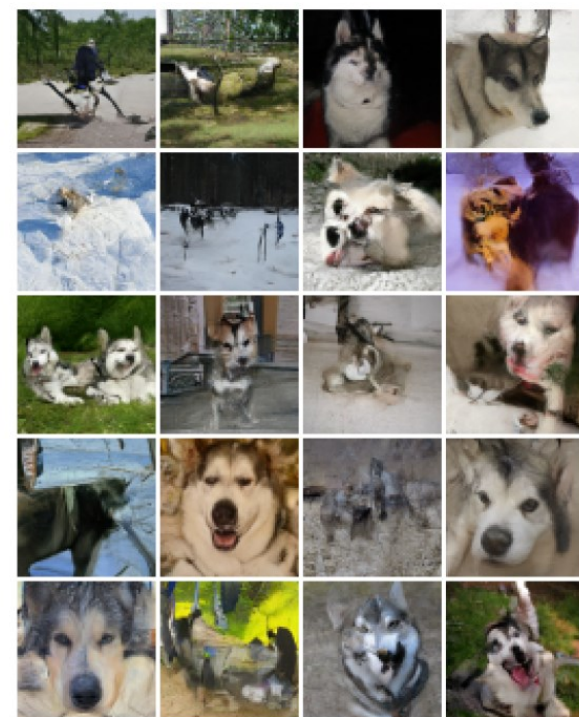
Random samples for 64x64 ImageNet



(a) 256 sampling steps



(b) 4 sampling steps



(c) 1 sampling step

On distillation of guided diffusion models

- Two-stage distillation:
 - 1) Distill unconditional and classifier-free-guidance DMs into a single one
 - Fourier embedding for w
 - 2) Progressively halve the step number through distillation until reaching 1-4 steps
 - DDIM distillation
 - Stochastic sampling w/ travel back



Text-guided generation (1 step)



Input Mask Result 1 Result 2

Image inpainting (2 steps)



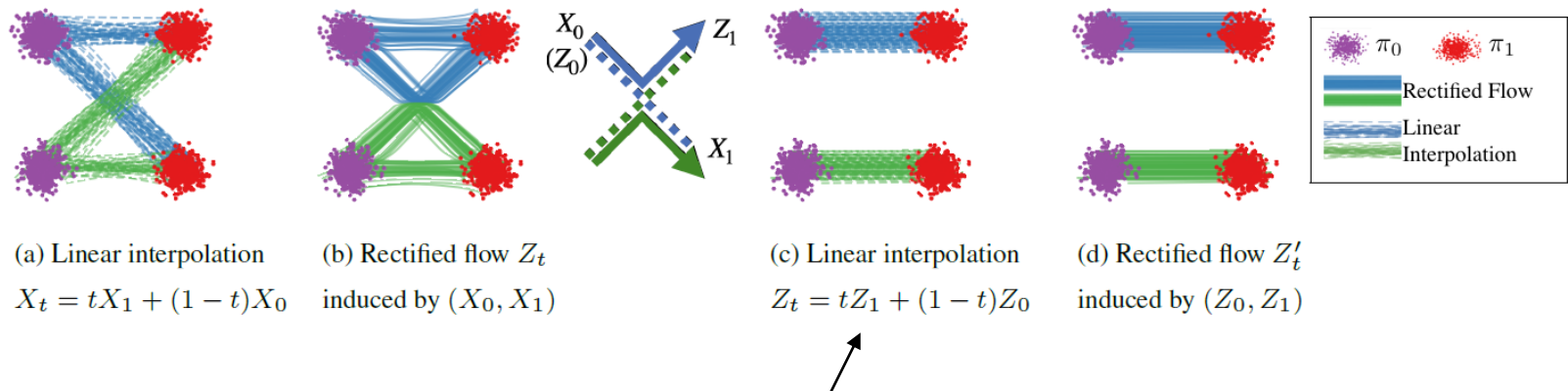
Input Output (different styles)

Image to image translation (3 steps)

Ref: 1. Dhariwal et al., NIPS'21; ADM. 2. Meng et al, CVPR'23; Distillation.

Reflow

- Multiple-pass rectification straightens the flow path
 - Simple and effective
 - Can further distill the k-rectified flow $Z_1^k \approx \hat{T}(Z_0^k)$



Sampling: Draw (Z_0, Z_1) following $dZ_t = v_{\hat{\theta}}(Z_t, t)dt$ starting from $Z_0 \sim \pi_0$ (or backwardly $Z_1 \sim \pi_1$)



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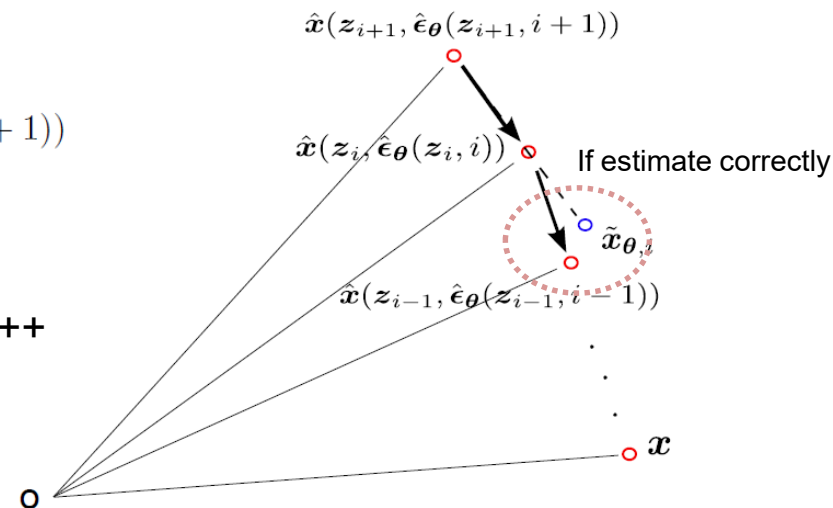


Training-free samplers

- Idea
 - A better sampler reduces discretization error for numeric integration and therefore decreases the number of steps
- Category
 - Mean/variance refinement, e.g.
 - LA-DPM [Zhang et al., CVPR'23]: lookahead update for mean estimation

$$\begin{aligned}\tilde{x}_{\theta,i}(\lambda_i) &= \hat{x}(z_i, \hat{\epsilon}_{\theta}(z_i, i)) + \lambda_i \Delta \hat{x}_{\theta,i} \\ &= (1 + \lambda_i) \hat{x}(z_i, \hat{\epsilon}_{\theta}(z_i, i)) - \lambda_i \hat{x}(\hat{\epsilon}_{\theta}(z_{i+1}, i + 1))\end{aligned}$$

- Fast samplers for ODE, e.g.
 - EDM, DPM-Solver, DEIS, DPM-Solver++
- Fast samplers for SDE, e.g.
 - SDE-DPM-Solver++, SA-Solver





Fast samplers for ODE (1/4)

- Observation

- Direct discretization (for reverse diffusion) leads to unnecessary error accumulation on the linear term

$$dx_t = \left(f_t x_t - \frac{g_t^2 s_\theta}{2} \right) dt \quad \xrightarrow{\substack{\text{Discretize on } t; \\ \text{Sample-and-hold on } f_i, g_i, \text{ and } s_\theta}} \quad x_{i+1} = \underbrace{x_i - f_i x_i \Delta t}_{\text{Error accumulation}} + \frac{g_i^2 s_\theta}{2} \Delta t$$

- Solution

- EDM [Karras et al., NIPS'22]
 - Eliminate the linear term with reparameterization
 - Solve non-linear term with 2nd-order Heun's method (two NFEs per step)
- DPM-Solver [Lu et al., NIPS'22]
 - Apply variation of parameters (VOP) for exact linear form (same as EDM reparam.)
 - Solve non-linear term with 2nd- or 3rd-order Runge-Kutta method (2 or 3 NFEs)
 - Sample on noise prediction model ϵ_θ
 - Discretize on log-SNR $\lambda_t = \log \alpha_t / \sigma_t$

$$\frac{dx_t}{dt} = h_\theta(x_t, t) := f(t)x_t + \frac{g^2(t)}{2\sigma_t} \epsilon_\theta(x_t, t) \quad \xrightarrow{\text{VOP}} \quad x_t = \underbrace{e^{\int_s^t f(\tau) d\tau} x_s}_{\text{No error}} + \underbrace{\int_s^t \left(e^{\int_\tau^t f(r) dr} \frac{g^2(\tau)}{2\sigma_\tau} \epsilon_\theta(x_\tau, \tau) \right) d\tau}_{\text{Analytically computable}}$$



Fast samplers for ODE (2/4)

- Diffusion Exponential Integrator Sampler (DEIS) [Zhang et al., ICLR'23]
 - Propose to solve the non-linear term with multi-step Adams-Bashforth (AB) method which applies high-order Lagrange polynomial extrapolation

$$\hat{\mathbf{x}}_{t_{i-1}} = \Psi(t_{i-1}, t_i) \hat{\mathbf{x}}_{t_i} + \sum_{j=0}^r [C_{ij} \epsilon_{\theta}(\hat{\mathbf{x}}_{t_{i+j}}, t_{i+j})]$$

Only one NFE required
for one step

$$C_{ij} = \int_{t_i}^{t_{i-1}} \frac{1}{2} \Psi(t_{i-1}, \tau) G_{\tau} G_{\tau}^T L_{\tau}^{-T} \prod_{k \neq j} \left[\frac{\tau - t_{i+k}}{t_{i+j} - t_{i+k}} \right] d\tau$$

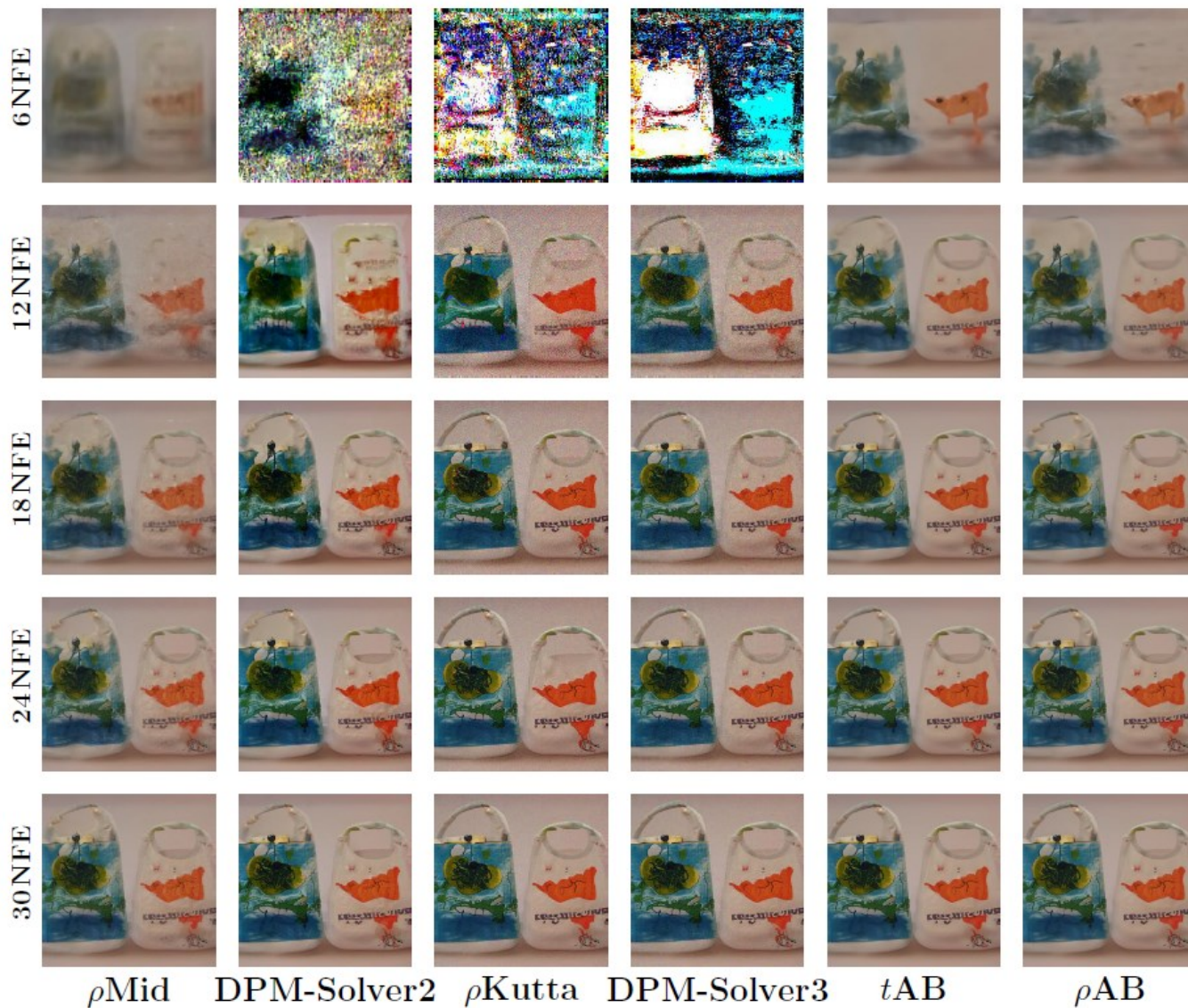
- Sample on noise prediction model ϵ_{θ}
- Discretize
 - on t for VOP-formulated semi-linear ODE
 - or on inverse-SNR $\rho_t = \sigma_t / \alpha_t$ for reparameterized non-linear ODE
- FID quality: $tAB > DPM\text{-}Solver(\lambda RK) > \rho AB > \rho RK > EDM$

Unconditional synthesis on ImageNet 256x256

NFE	FID for various DEIS									
	DDIM	$\rho 2Heun^{\dagger}$	$\rho 3Kutta$	$\rho 4RK$	$\rho AB1$	$\rho AB2$	$\rho AB3$	$tAB1$	$tAB2$	$tAB3$
5	26.91	108^{+1}	185^{+1}	193^{+3}	22.28	21.53	21.43	19.72	16.31	15.37
10	11.14	14.72	13.19^{+2}	28.65^{+2}	7.56	6.72	6.50	6.09	4.57	4.17
15	7.06	4.89^{+1}	5.88	6.88^{+1}	4.69	4.16	3.99	4.29	3.57	3.37
20	5.47	3.50	2.97^{+1}	3.92	3.70	3.32	3.17	3.54	3.05	2.86
50	3.27	2.60	2.55^{+1}	2.57^{+2}	2.70	2.62	2.59	2.67	2.59	2.57

10-step
is good
enough ←

Fast samplers for ODE (3/4)

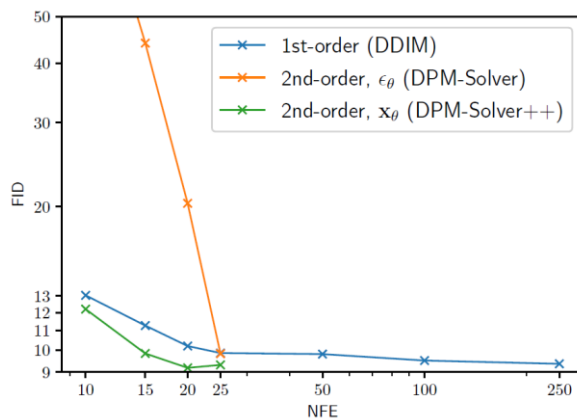




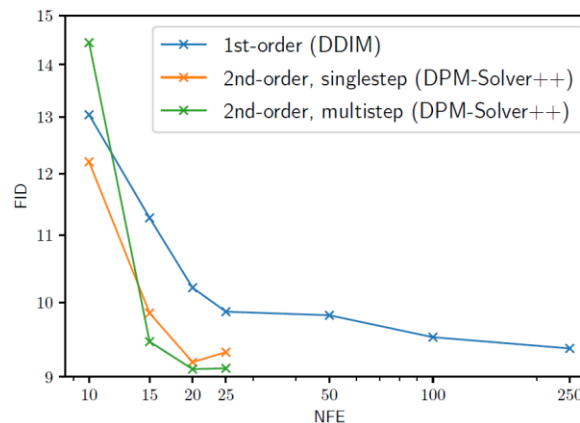
Fast samplers for ODE (4/4)

- DPM-Solver++ [Lu et al., arXiv'23] for guided synthesis
 - Sample on data prediction model x_θ , instead of ϵ_θ , for reducing discretization error for guided synthesis
 - Adopt multi-step method (through Taylor expansion and some simplification) to solve the non-linear term for high-order solver
 - DDIM $\equiv$ DPM-Solver1 (1st-order with $\epsilon_\theta(\lambda)$) $\equiv$ DPM-Solver++1 (1st-order with $x_\theta(\lambda)$)

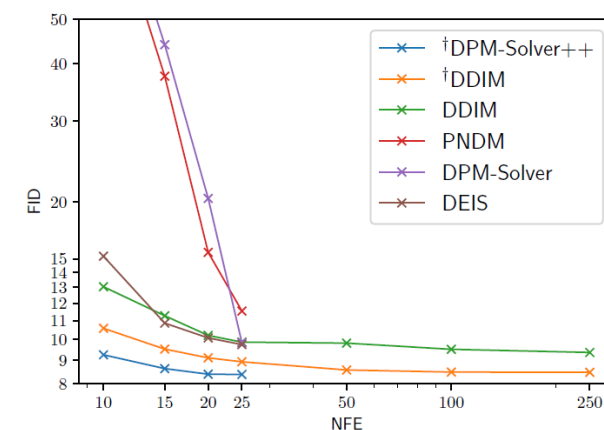
Guided synthesis with s=8 on ImageNet 256x256



Data model better than noise model



Multi-step is better when NFE>15



Thresholding[†] is important



Fast samplers for SDE (1/2)

- Solution
 - SDE-DPM-Solver++ [Lu et al., arXiv'23]
 - Extends data prediction model for SDE solving
 - DDPM $\equiv$ SDE-DPM-Solver++1 (1st-order with $x_\theta(\lambda)$)
 - SA-Solver [Xue et al., arXiv'23]
 - Stochastic Adams (SA) method with x_θ sampling and t discretization
 - Predictor (s -step): $x_{i+1}^p \leftarrow \{x_i^c, x_\theta(x_i^p), \dots, x_\theta(x_{i-(s-1)}^p), \xi(\omega_i) \sim N(\mathbf{0}, \mathbf{I})\}$ by Lagrange extrapolation
 - Corrector ($\hat{s}$ -step): $x_{i+1}^c \leftarrow \{x_i^c, x_\theta(x_{i+1}^p), \dots, x_\theta(x_{i-(\hat{s}-1)}^p), \xi(\omega_i)\}$ by Lagrange interpolation
 - One NFE, $x_\theta(x_{i+1}^p)$, for one predictor and corrector update
 - $x_\theta(x_{i+1}^p)$ is used to correct x_{i+1}^c for the same sample path of $\xi(\omega_i)$
 - Considers SDE family and gives the general formulation with stochasticity $\tau(t)$
 - $\tau(t) = 0$: ODE; $\tau(t) = 1$: original SDE; $\tau(t) > 1$: SDE with additional stochasticity

VOC with
log-SNR λ

$$dx_t = \left[f(t)x_t - \left(\frac{1 + \tau^2(t)}{2} \right) g^2(t) \nabla_x \log p_t(x_t) \right] dt + \tau(t)g(t)d\bar{w}_t$$

$$x_t = \underbrace{\frac{\sigma_t}{\sigma_s} e^{-\int_{\lambda_s}^{\lambda_t} \tau^2(\tilde{\lambda}) d\tilde{\lambda}} x_s}_{\text{exact}} + \underbrace{\sigma_t F_\theta(s, t)}_{\text{solver}} + \underbrace{\sigma_t G(s, t)}_{\text{exact}}$$

$$F_\theta(s, t) = \int_{\lambda_s}^{\lambda_t} e^{-\int_{\lambda_u}^{\lambda_t} \tau^2(\tilde{\lambda}) d\tilde{\lambda}} (1 + \tau^2(\lambda)) e^\lambda x_\theta(x_\lambda, \lambda) d\lambda$$

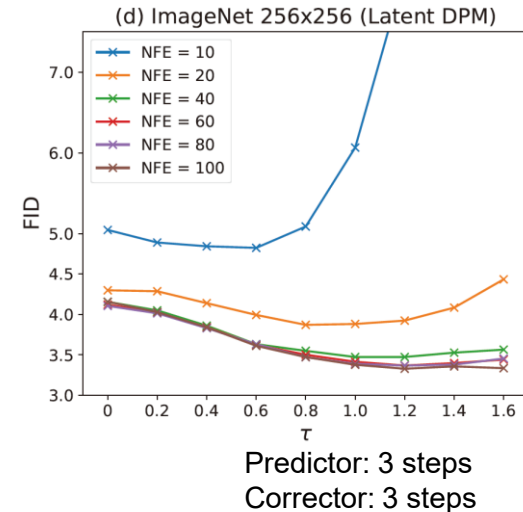
$$G(s, t) = \int_s^t e^{-\int_{\lambda_u}^{\lambda_t} \tau^2(\tilde{\lambda}) d\tilde{\lambda}} \tau(u) \sqrt{-2 \frac{d\lambda_u}{du}} d\bar{w}_u,$$



Fast samplers for SDE (2/2)

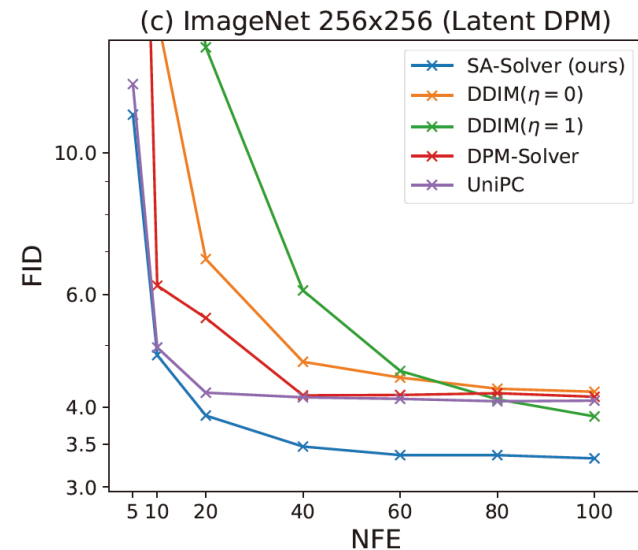
– SA-Solver (cont'd) [Xue et al., arXiv'23]

- Shows the equivalence between
 - General update rule of DDIM- η and 1-step SA-predictor
 - DPM-Solver++(2M) and 2-step SA-predictor with $\tau(t) = 0$
- Checks the effectiveness of the corrector and stochasticity $\tau(t)$
 - Small NFEs prefer small $\tau(t)$ (towards ODE)
 - Larger NFEs prefer large $\tau(t)$; stochasticity is effective in this case
 - Explain why DDIM is better than DDPM for small NFE: lack of correctors



VE, CIFAR10 32x32

method \ setting (NFE, τ)	15,0.4	23,0.8	31,1.0	47,1.4
Predictor 1-steps only	13.76	12.44	11.72	14.67
Predictor 1-steps, Corrector 1-step	8.49	6.87	6.13	6.75
Predictor 3-steps only	5.30	3.93	3.52	2.98
Predictor 3-steps, Corrector 3-steps	4.91	3.77	3.40	2.92





Summary of fast samplers

- SOTA achieves 10-step sampling with VOP-formulated ODE/SDE
- Characterized in four aspects
 - **Prediction**: noise model is popular for unconditional synthesis with ODE; data model is good for guided synthesis and SDE
 - **Discretization**: simple t-discretization seems good enough, although the formulations with log-SNR are more reasonable
 - **Solver**: multi-step shows quality advantages; corrector improves SDE
 - **ODE/SDE**: SDE can beat ODE with corrector and stochasticity

Characteristic	Prediction		Discretization			Solver		ODE/SDE	
	ϵ_θ	x_θ	ρ	t	λ	single-step (e.g. RK)	Multi-step (e.g. AB)	ODE	SDE
EDM		V		V		V		V	
DPM-Solver	V				V	V		V	
DEIS	V		V				V (P)	V	
DPM-Solver++		V			V		V (P)	V	
SDE-DPM-Solver++		V			V		V (P)		V
SA-Solver		V			V		V (PC)	V	V

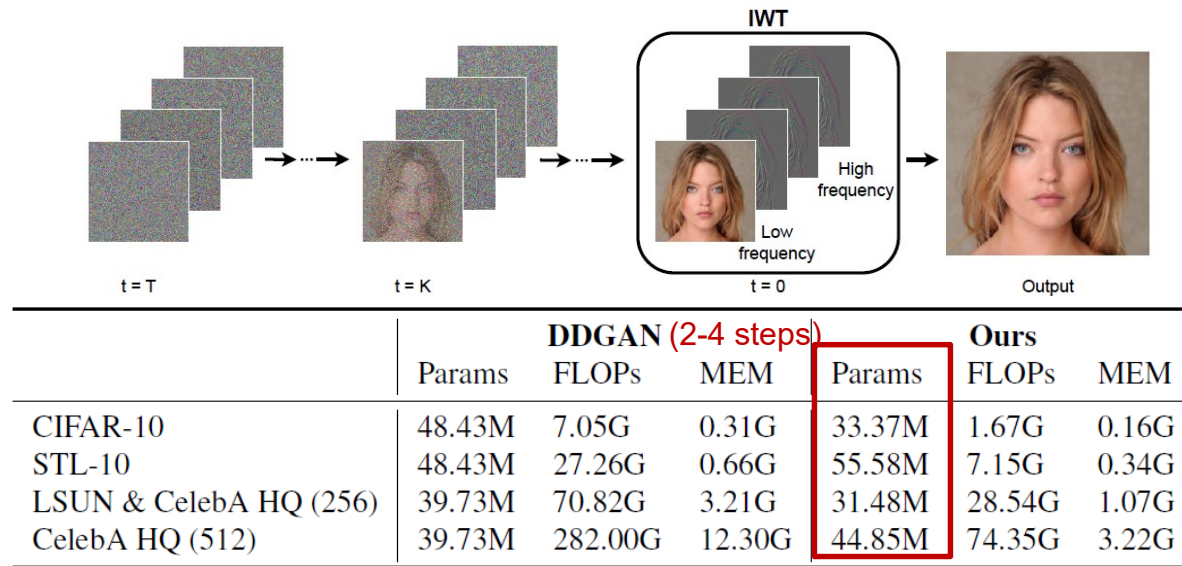


Outline

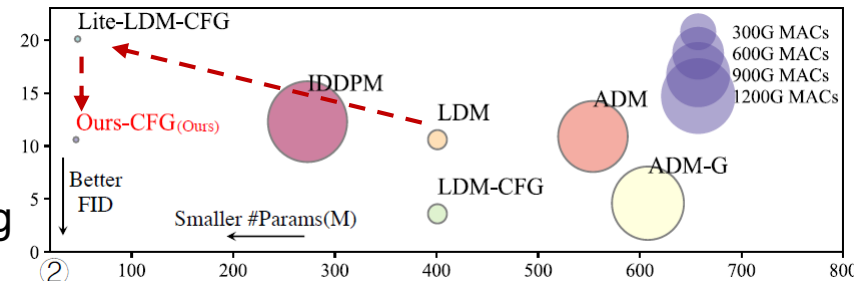
- **Implementation**
 - Image synthesis
 - Task-specific image restoration and enhancement
 - Zero-shot image restoration
 - Size-agnostic image generation
 - Video generation
 - **Diffusion acceleration**
 - Step reduction
 - **Model shrinkage**
 - **Model quantization**

Model shrinkage

- Wavelet diffusion model
 - Performs diffusion in wavelet sub-bands which are downsampled by 2
 - Proposes wavelet-embedded generator



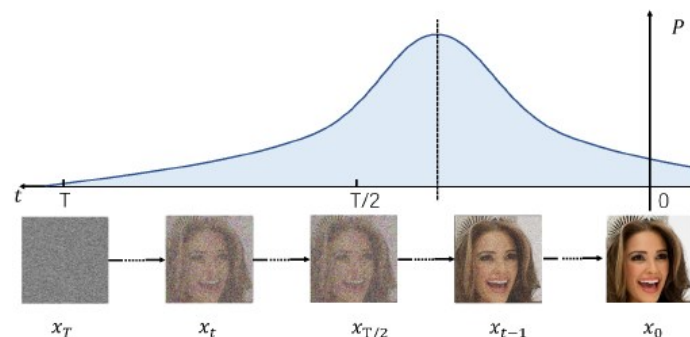
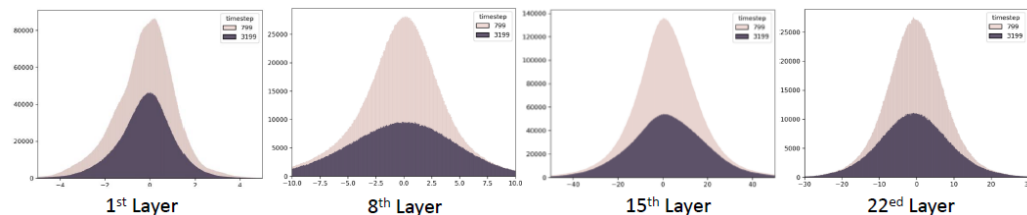
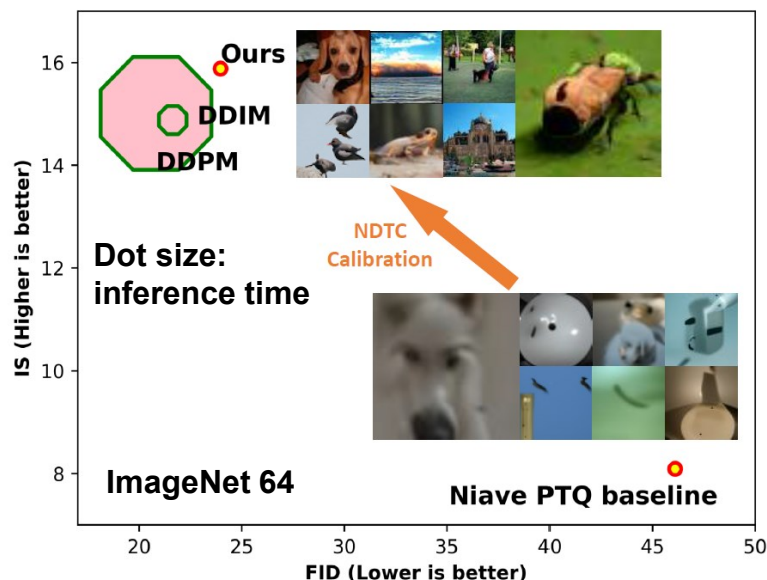
- Spectral diffusion model
 - ~10x reduction of model size from LDM
 - Spatial/spectral distillation, wavelet gating



Ref: 1. Phung et al., CVPR'23; Wavelet DM. 2. Yang et al, CVPR'23, Spectral DM.

Model quantization (1/2)

- Post-training quantization for diffusion models (PTQ4DM)
 - PTQ with layer-wise 8-bit W/A for Conv2d and linear layers
 - Weight: L2 calibration
 - Activation: normally distributed time-step calibration (NDTC)
 - Issue: activation distributions vary over time
 - Solution: collect calibration data over time-step sampling as biased Gaussian

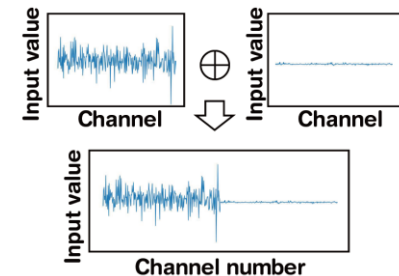


Ref: Shang et al., CVPR'23; PTQ4DM.



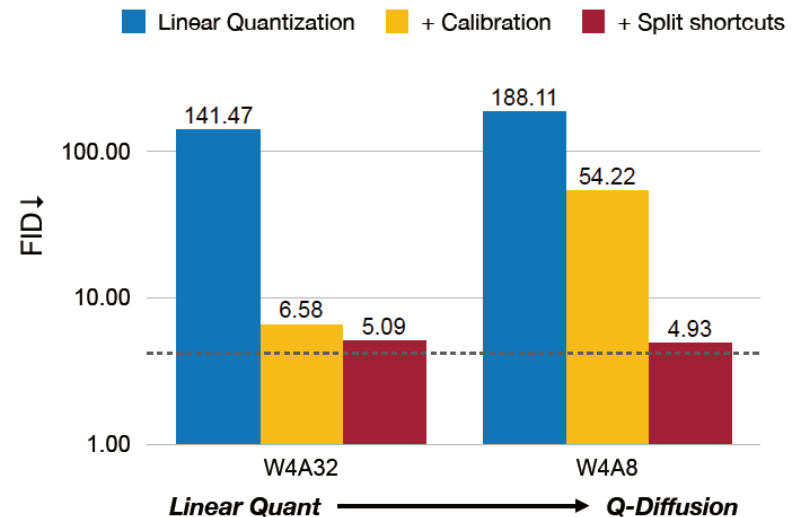
Model quantization (2/2)

- Q-Diffusion
 - PTQ with channel-wise 4-bit weight and layer-wise 8-bit activation
 - Includes attention and verifies large-res. results, e.g. Stable Diffusion
 - Weight: L2 calibration
 - Activation: uniformly sampled for calibration
 - Shortcut-split quantization (enables W4)



LDM-4 on LSUN-Bedrooms 256

Method	Bits (W/A)	Size (Mb)	TBops	FID↓
Full Precision	32/32	1096.2	107.17	2.98
Linear Quant	8/32	274.1	37.28	3.02
SQuant	8/32	274.1	37.28	2.94
Q-Diffusion	8/32	274.1	37.28	2.97
Linear Quant	4/32	137.0	18.64	82.69
SQuant	4/32	137.0	18.64	149.97
Q-Diffusion	4/32	137.0	18.64	4.86
Linear Quant	8/8	274.1	12.97	6.69
SQuant	8/8	274.1	12.97	4.92
Q-Diffusion	8/8	274.1	12.97	4.40
Linear Quant	4/8	137.0	6.48	24.86
SQuant	4/8	137.0	6.48	95.92
Q-Diffusion	4/8	137.0	6.48	5.32



Ref: Li et al., ICCV'23; Q-Diffusion.



Conclusion

- DM opens up new avenues for image processing tasks
 - Realistic image synthesis and editing
 - Superior image restoration and enhancement
 - General processing framework
- DM has CT-DT duality
 - CT SDE/ODE enables portability and gives insights
 - DT generative process serves as numerical integration of SDE
- DM acceleration is on the way (at a fast pace)
 - FM enables 10-25 step generation
 - Distillation and reflow enable 1-4 step generation
 - Fast sampling enables 10-step generation
 - Latent DM and distillation reduce model size
 - Transformer structure (DiT) reduces computation complexity
 - Block floating-point formats resolve quantization issues

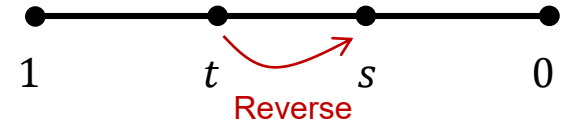


Outline

- **Implementation**
 - Image synthesis
 - Task-specific image restoration and enhancement
 - Zero-shot image restoration
 - Size-agnostic image generation
 - Video generation
 - Diffusion acceleration
 - **Appendices for equations**
 - **v-prediction**



v-prediction



- Formulation of DDIM sampler for VP SDE ($\alpha_t^2 + \sigma_t^2 = 1$)
 - Prediction: $\mathbf{x}_t = \alpha_t \mathbf{x}_\theta^{(t)} + \sigma_t \epsilon_\theta^{(t)}$ which predicts data $\mathbf{x}_0$ by $\mathbf{x}_\theta^{(t)}$ and noise ϵ_t by $\epsilon_\theta^{(t)}$
 - Generation: $\mathbf{x}_s = \alpha_s \mathbf{x}_\theta^{(t)} + \sigma_s \epsilon_\theta^{(t)}$
- Unstable issues for conventional generation samplers
 - Noise-based $\mathbf{x}_s = \alpha_s \frac{\mathbf{x}_t - \sigma_t \epsilon_\theta^{(t)}}{\alpha_t} + \sigma_s \epsilon_\theta^{(t)}$ unstable when $\alpha_t \rightarrow 0$ as $t \rightarrow 1$
 - Data-based $\mathbf{x}_s = \alpha_s \mathbf{x}_\theta^{(t)} + \sigma_s \frac{\mathbf{x}_t - \alpha_t \mathbf{x}_\theta^{(t)}}{\sigma_t}$ unstable when $\sigma_t \rightarrow 0$ as $t \rightarrow 0$
- Stabilization with v-prediction
 - $\mathbf{v}_\theta^{(t)} := \alpha_t \epsilon_\theta^{(t)} - \sigma_t \mathbf{x}_\theta^{(t)}$; equivalence: $\epsilon_\theta^{(t)} = \alpha_t \mathbf{v}_\theta^{(t)} + \sigma_t \mathbf{x}_t$, $\mathbf{x}_\theta^{(t)} = \alpha_t \mathbf{x}_t - \sigma_t \mathbf{v}_\theta^{(t)}$
 - $\mathbf{x}_s = (\alpha_s \alpha_t + \sigma_s \sigma_t) \mathbf{x}_t + (\alpha_t \sigma_s - \alpha_s \sigma_t) \mathbf{v}_\theta^{(t)}$ which is always stable without division
- Viewpoint of angular parameterization [T. Salimans et al., ICLR'22]
 - $\phi_t := \arctan \frac{\sigma_t}{\alpha_t} \Rightarrow \alpha_t = \cos \phi_t, \sigma_t = \sin \phi_t$
 - $\mathbf{v}_\theta^{(t)} = \frac{\partial \mathbf{x}_t}{\partial \phi_t}$ which predicts the change rate over ϕ_t for $\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \epsilon_t$ (forward process)
 - $\mathbf{x}_s = \cos(\phi_s - \phi_t) \mathbf{x}_t + \sin(\phi_s - \phi_t) \mathbf{v}_\theta^{(t)}$ (DDIM)